

Debiased Inference for Bounding Wage Inequality with Many Controls¹

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Summary: We study estimation and inference for a partially identified parameter whose identified set depends on a first-stage nuisance parameter that must itself be estimated. Combining the criterion-function approach with the theory of Neyman-orthogonal moments that underlies double/debiased machine learning, we propose a two-step procedure: the point-identified nuisance is estimated by flexible machine-learning methods, and the set-identified target is recovered as a level set of a sample criterion built from orthogonal moment inequalities with cross-fitting. When the contour level is bounded, we show that the resulting set estimator converges in Hausdorff distance at the parametric rate of the infeasible criterion built on the true nuisance. We further develop a subsampling procedure that delivers asymptotically valid coverage, provided the product of the first-stage estimation errors is $o(N^{-1/2})$. We illustrate the method on bounds for the wage distribution and the interquantile range under selection into employment and on the gender wage gap with an interval-censored wage. The empirical application studies the gender wage gap using the March supplement of the 2015 Current Population Survey.

Keywords: *Bounds on wage distributions; bracketed (interval-valued) data; sample selection; double/debiased machine learning; subsampling; partial identification; moment inequalities.*

1. INTRODUCTION

Sample selection is a classic problem in econometrics (Heckman, 1974). Economically important objects such as population wage distributions, gender and education wage differentials, and measures of wage inequality depend on the wages of individuals who do not work, and these wages are not observed. One approach imposes enough structure on the joint distribution of wages and employment to recover the latent wage distribution, for instance through a control function. For example, Chernozhukov et al. (2025) take this route with distribution regression to decompose British wages. An alternative approach leaves the selection process incompletely specified and asks what can be learned about the latent wage distribution under weaker restrictions, as in the partial identification analysis of Manski (2003). In this paper, we focus on Blundell et al. (2007), who take the second route to bound changes in the distribution of British wages. Their positive-selection restriction, under which the wage distribution of workers first-order stochastically dominates that of nonworkers conditional on observed characteristics, substantially tightens

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the worst-case bounds without point-identifying the missing wage distribution. The restriction is more plausible when the compared individuals are observationally similar, because it rules out selection only within narrow groups. Conditioning on many characteristics, however, creates a dimensionality problem.

This paper develops estimation and inference for partially identified models whose identifying inequalities depend on a high-dimensional first stage. The parameter of interest θ is set-identified by a finite collection of unconditional moment inequalities,

$$Eg(Z; \theta, \xi_0) \leq 0,$$

where the nuisance parameter ξ_0 is point-identified but may be infinite-dimensional. We estimate ξ_0 by machine-learning methods with cross-fitting, construct Neyman-orthogonal versions of the inequalities, and estimate the identified set as a level set of the criterion function of Chernozhukov et al. (2007). We illustrate it with the wage-distribution bounds of Blundell et al. (2007) and a regression with an interval-valued outcome. The object of inference is the identified set with the confidence region covering Θ_I with probability approaching $1 - \tau$, where $1 - \tau$ is the nominal confidence level.

The main technical contribution is subsampling inference that remains valid when the criterion is built from a cross-fitted first stage. Cross-fitting breaks the independence across subsamples that subsampling needs, because every block statistic uses a nuisance estimated on the whole sample and is maximized over a preliminary set built from the whole sample. We restore it using a carefully crafted balanced, disjoint-block construction together with a bracketing argument. Every block contains the same number of observations from each cross-fitting fold, so the observations it draws from fold k are independent of the nuisance estimated without that fold. Each feasible block statistic is then bounded by statistics evaluated at the true nuisance on a small enlargement or contraction of the identified set. These bounding statistics depend only on the observations of their own block and can therefore be treated with ordinary block-level asymptotics. This yields consistency of the subsampling critical value and pointwise asymptotic coverage,

$$P\left(\Theta_I \subseteq C_N(\hat{c}_\tau; \hat{\xi})\right) = 1 - \tau + o(1),$$

under $b \rightarrow \infty$ and $bc_N/N \rightarrow 0$, with no strengthening of the first-stage rate condition used for set estimation. Combining cross-fitting with dependent data raises a related difficulty, resolved by a cross-fitting scheme tailored to the dependence, as in the weakly dependent panels of Semenova et al. (2023), or by an additional stability condition, as in the network and spatial designs of Cao and Leung (2025).

A second contribution is a general theory that does not require the moment inequalities to be affine in θ . Bounds on the interquantile range of the wage distribution under selection provide such an example.

We demonstrate our method on the partially identified model studied in Chandrasekhar et al. (2012), with wage data from the March supplement of the 2015 Current Population Survey. In particular, we document that, unlike an approach that reports only separate bounds on individual coefficients, the criterion-function formulation directly delivers a joint confidence region for the two-dimensional coefficient set. Across the designs we consider, the estimated joint identified set occupies only about 50% of the box formed by its coordinate projections. We interpret this as evidence that separate coefficient bounds contain many combinations that are inconsistent with the joint directional restrictions.

This paper contributes to the literature on partial identification by incorporating a

function-valued first stage into the two-stage framework of Kaido and White (2014). In doing so, we benefit from the Neyman-orthogonality and cross-fitting machinery of Chernozhukov et al. (2018), Chernozhukov et al. (2022) and Chernozhukov et al. (2022a), so that a high-dimensional first stage can be estimated by regularized methods without affecting first-order inference. In partial identification, earlier applications of these techniques are mostly confined to bounds on one-dimensional parameters (Heiler, 2024; Semenova, 2025) and to support functions of convex identified sets (Semenova, 2023; Liu and Molinari, 2024), building on Beresteanu and Molinari (2008), Beresteanu et al. (2011) and Bontemps et al. (2012). Liu and Molinari (2024), for instance, propose debiased estimation and inference for an algorithmic fairness-accuracy frontier, and Liu (2025) develops a partial identification framework for panel-data policy evaluation that is robust to violations of the identifying assumptions of difference-in-differences and synthetic control. In contrast, our moment inequalities may be nonlinear in θ and describe potentially non-convex identified sets. Last but not least, the paper’s emphasis on selection problems contributes to the sample selection literature, see Fernández-Val et al. (2023, 2024); Fernández-Val and Hong (2025).

Since the first version of this paper¹ (arXiv:1808.02569, August 2018), Tian (2025) has studied a closely related two-stage problem. He combines the refined conditional chi-squared test of Cox and Shi (2023) with a correction that propagates first-stage estimation uncertainty into the covariance matrix and inverts the test point by point, so that his confidence set covers each θ in the identified set with asymptotic probability at least the nominal level. The correction relies on an asymptotically linear first stage, such as GMM or maximum likelihood, and on a high-level condition that the plug-in moment vector be $\sqrt{N}$ -asymptotically normal, which a machine-learning first stage that converges more slowly does not deliver without orthogonalization. We instead remove the first-order contribution of the estimated nuisance by orthogonalization, allow a first stage that converges more slowly than $N^{-1/2}$, and cover Θ_I as a whole. The two procedures thus treat the first stage differently: Tian (2025) carries its first-order sampling variation into inference, while our construction makes that variation second order under suitable rate conditions.

Structure of the paper. Section 2 introduces the two-stage framework, the orthogonal criterion, and the estimation and inference procedure. Section 3 presents the two examples: the wage distribution under selection of Blundell et al. (2007), and the gender wage gap with an interval-valued outcome. Section 4 gives primitive conditions and asymptotic guarantees for both examples, and Section 5 reports the empirical application. The appendix, intended for online publication, collects the general asymptotic theory, the proofs, and the verification of the high-level conditions for the two examples.

2. SETUP

2.1. Framework

Let $Z_1, \dots, Z_N$ be an i.i.d. sample from a distribution P_0 on a measurable space $\mathcal{Z}$. The *target* parameter θ_0 lies in a compact set $\Theta \subset \mathbb{R}^{d_\theta}$ and is the object of economic interest. The *nuisance* parameter ξ lies in a convex subset Ξ of a normed space of functions of $\mathcal{Z}$,

¹Appendix A and its proofs, except the extension to moments that are discontinuous in θ , first appeared in Chapter 4 of arXiv:1808.02569, formerly titled “Machine Learning for Dynamic Discrete Choice.”

with the $L^2(P_0)$ norm $\|\cdot\|_{P,2}$ unless an example specifies another norm. The nuisance has a true value $\xi_0 \in \Xi$ that can be estimated without reference to θ_0 . Following Chernozhukov et al. (2018), we assume that all objects are measurable and focus on estimation and inference. The pair (θ_0, ξ_0) has the two-stage structure of Kaido and White (2014), with a point-identified first stage. Unlike their setting, the first stage here is an (infinite-dimensional) nuisance function rather than a Euclidean sub-vector.

A moment function $g : \mathcal{Z} \times \Theta \times \Xi \rightarrow \mathbb{R}^L$ defines the identified set through $L \geq 1$ inequalities evaluated at the true nuisance,

$$\Theta_I := \{\theta \in \Theta : \mathbb{E}g(Z; \theta, \xi_0) \leq 0\}, \quad (2.1)$$

where $v \leq 0$ for a vector $v \in \mathbb{R}^L$ means that every coordinate of v is non-positive. Write $\|v\|_+ := \|\max\{v, 0\}\|$ for the Euclidean norm of the positive part of v , the maximum being taken coordinatewise, so that $\|v\|_+ = 0$ if and only if $v \leq 0$. Define the *population criterion*

$$Q(\theta, \xi) := \|\mathbb{E}g(Z; \theta, \xi)\|_+^2 \quad (2.2)$$

This criterion is non-negative for every $\theta \in \Theta$ and $\xi \in \Xi$. At the true value of the nuisance, it vanishes exactly on the identified set:

$$\Theta_I = \{\theta \in \Theta : Q(\theta, \xi_0) = 0\}.$$

We construct the sample criterion function by K -fold cross-fitting, with the number of folds $K \geq 2$ fixed. The indices $\{1, \dots, N\}$ are partitioned into folds $J_1, \dots, J_K$ of equal size $n = N/K$, and for $i \in J_k$ the nuisance $\hat{\xi}_i := \hat{\xi}^{(k)}$ is estimated from the observations outside J_k only. We write $\hat{\xi}$ for the collection $(\hat{\xi}^{(1)}, \dots, \hat{\xi}^{(K)})$. The *sample criterion* is

$$Q_N(\theta, \hat{\xi}) := \left\| \frac{1}{N} \sum_{i=1}^N g(Z_i; \theta, \hat{\xi}_i) \right\|_+^2, \quad (2.3)$$

and for a deterministic $\xi \in \Xi$ we write $Q_N(\theta, \xi)$ for the same expression with ξ in place of every $\hat{\xi}_i$. At $\xi = \xi_0$ it is the criterion of Chernozhukov et al. (2007) for these moment inequalities. For a level $c \geq 0$ and a nuisance ξ , the contour set of the sample criterion is

$$C_N(c; \xi) := \{\theta \in \Theta : N Q_N(\theta, \xi) \leq c\}. \quad (2.4)$$

DEFINITION 2.1. (TWO-STAGE SET ESTIMATOR) *For a measurable level $\hat{c} \geq 0$, the two-stage set estimator is*

$$\hat{\Theta}_I := C_N(\hat{c}; \hat{\xi}). \quad (2.5)$$

The level $\hat{c}$ is chosen in one of two ways. A deterministic level $c_N \rightarrow \infty$ with $c_N/N \rightarrow 0$, for instance $c_N = \log N$, makes $\hat{\Theta}_I$ a consistent set estimator. A data-driven level $\hat{c}_\tau$, computed by subsampling, makes it a confidence region for Θ_I with asymptotic coverage $1 - \tau$.

DEFINITION 2.2. (SUBSAMPLING CRITICAL VALUE) *Fix a deterministic level $c_N \rightarrow \infty$ with $c_N/N \rightarrow 0$ and a block size $b = b_N \rightarrow \infty$, a multiple of K , with $b c_N/N \rightarrow 0$. Partition the sample into $B_N := \lfloor N/b \rfloor$ disjoint blocks $\mathcal{I}_1, \dots, \mathcal{I}_{B_N}$ of size b , each containing b/K observations from every fold, so that the observations of a block in fold k*

are independent of $\widehat{\xi}^{(k)}$. For each block $j = 1, \dots, B_N$ compute

$$\widehat{T}_{j,b} := \sup_{\theta \in C_N(c_N; \widehat{\xi})} b Q_{j,b}(\theta, \widehat{\xi}), \quad Q_{j,b}(\theta, \widehat{\xi}) := \left\| \frac{1}{b} \sum_{i \in \mathcal{I}_j} g(Z_i; \theta, \widehat{\xi}_i) \right\|_+^2,$$

so that $\widehat{T}_{j,b}$ is the criterion of block $\mathcal{I}_j$, scaled by b and maximized over the preliminary set $C_N(c_N; \widehat{\xi})$. Let $\widehat{G}$ be the empirical distribution function of $(\widehat{T}_{j,b})_{j=1}^{B_N}$. For $\tau \in (0, 1)$, the subsampling critical value is the $(1 - \tau)$ -quantile

$$\widehat{c}_\tau := \inf\{x \in \mathbb{R} : \widehat{G}(x) \geq 1 - \tau\}.$$

Rates for the set estimator are stated in the Hausdorff metric,

$$d_H(A, B) := \max \left\{ \sup_{x \in A} d(x, B), \sup_{y \in B} d(y, A) \right\}, \quad d(x, B) := \inf_{y \in B} \|x - y\|, \quad (2.6)$$

and $\Theta_I^\epsilon := \{\theta \in \Theta : d(\theta, \Theta_I) \leq \epsilon\}$ denotes the ϵ -expansion of the identified set.

The first stage enters (2.3) through $\widehat{\xi}$, whose estimation error is of larger order than $N^{-1/2}$ when ξ_0 is estimated by regularized methods. We require the moment function to be insensitive to that error to first order. For $\xi \in \Xi$ and $r \in [0, 1]$ write $\xi_r := \xi_0 + r(\xi - \xi_0)$, which lies in Ξ .

DEFINITION 2.3. (NEYMAN ORTHOGONALITY) *The moment function g is Neyman-orthogonal at ξ_0 on a set $\mathcal{N} \subset \Xi$ if, for every $\theta \in \Theta$ and $\xi \in \mathcal{N}$, the map $r \mapsto \mathbb{E} g(Z; \theta, \xi_r)$ is differentiable at $r = 0$ with*

$$\partial_r \mathbb{E} g(Z; \theta, \xi_r) \big|_{r=0} = 0. \quad (2.7)$$

Orthogonality says that, at the truth, the expected moment is flat in the direction of any first-stage error, so an error in $\widehat{\xi}$ moves $\mathbb{E} g$ only to second order.

2.2. Proposed procedure

The following algorithm combines the cross-fitted first stage, the criterion (2.3) and the contour sets (2.4) of Section 2.1 with a subsampling choice of the level. The orthogonal moment function g is an input, supplied case by case. Step 1 is the sample splitting of Schick (1986), in the cross-fitted form of Chernozhukov et al. (2018), and Steps 2–4 are the contour-set estimator and the subsampling procedure of Chernozhukov et al. (2007) applied to the cross-fitted criterion. Step 3 incorporates the proposed delicate combination of subsampling with cross-fitting.

ALGORITHM 2.1. (TWO-STAGE SET ESTIMATOR AND CONFIDENCE REGION) *Inputs: an orthogonal moment function g with nuisance ξ , the number of folds $K \geq 2$, a preliminary level $c_N \rightarrow \infty$ with $c_N/N \rightarrow 0$, a block size $b \rightarrow \infty$, a multiple of K , with $b c_N/N \rightarrow 0$, and a nominal confidence level $1 - \tau$.*

- 1 Cross-fitted first stage. Partition $\{1, \dots, N\}$ into folds $J_1, \dots, J_K$ of equal size $n = N/K$. For each k , compute an estimate $\widehat{\xi}^{(k)}$ of ξ_0 from the observations outside J_k , and for $i \in J_k$ set $\widehat{\xi}_i := \widehat{\xi}^{(k)}$.

2 Criterion and contour sets. Form the cross-fitted criterion (2.3) and the contour sets (2.4),

$$Q_N(\theta, \hat{\xi}) = \left\| \frac{1}{N} \sum_{i=1}^N g(Z_i; \theta, \hat{\xi}_i) \right\|_+^2, \quad C_N(c; \hat{\xi}) = \{\theta \in \Theta : N Q_N(\theta, \hat{\xi}) \leq c\}.$$

3 Preliminary set and block statistics. Form the preliminary set $C_N(c_N; \hat{\xi})$ at the deterministic level c_N , for instance $c_N = \log N$. Partition the sample into $B_N = \lfloor N/b \rfloor$ disjoint blocks $\mathcal{I}_1, \dots, \mathcal{I}_{B_N}$ of size b , each carrying b/K observations from every fold. On block j compute

$$Q_{j,b}(\theta, \hat{\xi}) = \left\| \frac{1}{b} \sum_{i \in \mathcal{I}_j} g(Z_i; \theta, \hat{\xi}_i) \right\|_+^2, \quad \hat{\mathcal{T}}_{j,b} = \sup_{\theta \in C_N(c_N; \hat{\xi})} b Q_{j,b}(\theta, \hat{\xi}),$$

4 Level and region. Let $\hat{G}$ be the empirical distribution function of $(\hat{\mathcal{T}}_{j,b})_{j \leq B_N}$ and let

$$\hat{c}_\tau := \inf \{x : \hat{G}(x) \geq 1 - \tau\}$$

be its $(1 - \tau)$ -quantile, as in Definition 2.2. Report the set estimator $\hat{\Theta}_I = C_N(\hat{c}; \hat{\xi})$ of Definition 2.1 with $\hat{c} = c_N$, or with $\hat{c} = 0$ when the sample criterion vanishes on a set with non-empty interior, together with the confidence region $C_N(\hat{c}_\tau; \hat{\xi})$.

In applications, the supremum in Step 3 can be computed on a finite grid. The preliminary set is represented by the grid points at which $N Q_N(\theta, \hat{\xi}) \leq c_N$, with spacing small relative to $\sqrt{c_N/N}$, and each block statistic is subsequently maximized over these points. When g is affine in θ , the block criterion and the preliminary set are convex, so only boundary grid points need to be evaluated.

3. EXAMPLES

3.1. Wage distribution under selection into employment

Our first example is the wage-distribution problem of Blundell et al. (2007). They study how the distribution of UK log hourly wages changed between 1978 and 2000 using the Family Expenditure Survey, and they report the interquartile range as a measure of within-group inequality together with educational, gender and cohort differentials in the median. Because wages are recorded only for active workers, these objects are only partially identified, and Blundell et al. (2007) bound them under restrictions on selection into work.

3.1.1. Linear case: the distribution function on a grid For each individual $i = 1, \dots, N$, let W_i denote the log wage, $D_i \in \{0, 1\}$ an employment indicator, and $X_i \in \mathbb{R}^p$ a vector of conditioning covariates. Selection into employment is non-ignorable, $W \not\perp D \mid X$, and we impose no exclusion restriction. Fix a grid of wage thresholds $w_1 < w_2 < \dots < w_S$. The target parameter is the vector of population distribution-function values on the grid,

$$\theta_0 := (F_0(w_1), \dots, F_0(w_S))', \quad F_0(w_k) := P(W \leq w_k), \quad k = 1, \dots, S. \quad (3.1)$$

Write $F_0(w \mid x) := P(W \leq w \mid X = x)$ for the conditional distribution function, $F_0^1(w \mid x)$ and $F_0^0(w \mid x)$ for its counterparts among the employed and the unemployed,

and $p_0(x) := P(D = 1 \mid X = x)$ for the *employment propensity*. The law of iterated expectations gives

$$F_0(w \mid x) = F_0^1(w \mid x)p_0(x) + F_0^0(w \mid x)(1 - p_0(x)). \quad (3.2)$$

The propensity $p_0(x)$ and the employed-wage distribution $F_0^1(w \mid x)$ are identified from the data (provided that $p_0(x) > 0$). On the contrary, the unemployed-wage distribution $F_0^0(w \mid x)$ is not identified, and it is restricted only by $F_0^0(w \mid x) \in [0, 1]$. Substituting into (3.2) gives the worst-case bounds of Manski (1994),

$$F_0^1(w \mid x)p_0(x) \leq F_0(w \mid x) \leq F_0^1(w \mid x)p_0(x) + 1 - p_0(x),$$

whose width $1 - p_0(x)$ is the non-employment rate for a given value of x . Blundell et al. (2007) tighten the lower bound under positive selection into employment, that is, first-order stochastic dominance of the employed wage distribution, $F_0^1(w \mid x) \leq F_0^0(w \mid x)$. Under that restriction it holds that

$$F_0^1(w \mid x) \leq F_0(w \mid x) \leq F_0^1(w \mid x)p_0(x) + (1 - p_0(x)). \quad (3.3)$$

To convert the problem to unconditional moment inequalities framework of Section 2, we integrate the bounds in (3.3) over the distribution of X , resulting in the bounds for each coordinate of θ_0 ,

$$\ell_{0,k} := E[F_0^1(w_k \mid X)], \quad u_{0,k} := E[F_0^1(w_k \mid X)p_0(X) + 1 - p_0(X)], \quad (3.4)$$

which gives

$$\ell_{0,k} \leq \theta_{0,k} \leq u_{0,k}.$$

A second restriction, that $F_0^0(\cdot \mid x)$ is a distribution function, ties the coordinates together. Applying (3.2) at $w_j < w_k$ and subtracting gives $F_0(w_k \mid x) - F_0(w_j \mid x) \geq p_0(x)(F_0^1(w_k \mid x) - F_0^1(w_j \mid x))$, since $F_0^0(\cdot \mid x)$ is non-decreasing. Integrating over X and using

$$E[p_0(X)(F_0^1(w_k \mid X) - F_0^1(w_j \mid X))] = E[D \mathbf{1}\{w_j < W \leq w_k\}]$$

turns this into a restriction on the coordinates of θ_0 ,

$$\theta_{0,k} - \theta_{0,j} \geq E[D \mathbf{1}\{w_j < W \leq w_k\}] = u_{0,k} - u_{0,j}, \quad j < k. \quad (3.5)$$

Combining (3.4) and (3.5) gives the identified set

$$\Theta_I = \left\{ \theta \in \Theta : \ell_{0,k} \leq \theta_k \leq u_{0,k} \quad \forall k, \quad \theta_k - \theta_{k-1} \geq u_{0,k} - u_{0,k-1} \quad \forall k \geq 2 \right\}, \quad (3.6)$$

a convex polytope cut out by $3S - 1$ inequalities.

In the notation of Section 2, the observation is $Z_i := (X_i, D_i, D_i W_i)$, $i = 1, \dots, N$, and the moment function has $3S - 1$ coordinates. For $k = 1, \dots, S$ the two coordinate moments are $g_k^{\text{lo}}(Z; \theta, \xi) := \phi_L(Z; w_k, \xi) - \theta_k$ and $g_k^{\text{up}}(Z; \theta) := \theta_k - \phi_U(Z; w_k)$, where

$$\phi_L(Z; w, \xi) := F^1(w \mid X) + \frac{D(\mathbf{1}\{W \leq w\} - F^1(w \mid X))}{p(X)}, \quad (3.7)$$

$$\phi_U(Z; w) := D \mathbf{1}\{W \leq w\} + 1 - D,$$

and for $k \geq 2$ the increment moments are

$$g_k^{\text{inc}}(Z; \theta) := \mathbf{1}\{w_{k-1} < W \leq w_k\}D - (\theta_k - \theta_{k-1}). \quad (3.8)$$

Collecting $\{g_k^{\text{lo}}, g_k^{\text{up}}\}_{k=1}^S$ and $\{g_k^{\text{inc}}\}_{k=2}^S$ gives a moment function $g(Z; \theta, \xi)$ of length $L =$

$3S - 1$, and the identified set (2.1) for θ_0 is (3.6). The first-stage nuisance parameter is $\xi_0 := (p_0, F_0^1(w_1 | \cdot), \dots, F_0^1(w_S | \cdot))$, estimated by cross-fitting.

Note that even in the linear case, it is necessary construct the orthogonal score ϕ_L as in (3.7). The two plug-in estimators of the lower bound $\ell_{0,k}$, the regression form $N^{-1} \sum_i \hat{F}^1(w_k | X_i)$ and the weighting form $N^{-1} \sum_i D_i \mathbf{1}\{W_i \leq w_k\} / \hat{p}(X_i)$, inherit the first-stage error to first order. For a nuisance value $\xi = (p, F^1)$,

$$\begin{aligned} E[F^1(w_k | X)] - \ell_{0,k} &= E[F^1(w_k | X) - F_0^1(w_k | X)], \\ E\left[\frac{D \mathbf{1}\{W \leq w_k\}}{p(X)}\right] - \ell_{0,k} &= E\left[F_0^1(w_k | X) \frac{p_0(X) - p(X)}{p(X)}\right], \\ E\phi_L(Z; w_k, \xi) - \ell_{0,k} &= E\left[\left(F^1(w_k | X) - F_0^1(w_k | X)\right) \frac{p(X) - p_0(X)}{p(X)}\right]. \end{aligned} \quad (3.9)$$

The first two errors are linear in a single first-stage error, typically of larger order than $N^{-1/2}$, so coverage of Θ_I is no longer guaranteed. The third error is the product of the two first-stage errors, which is $o(N^{-1/2})$ when each converges faster than $N^{-1/4}$. Every inequality in (3.6) is affine in θ , so the orthogonal estimates of $\ell_{0,k}$ and $u_{0,k}$ could also be potentially used in the standard moment-inequality inference on the estimated system (Andrews and Soares, 2010; Cox and Shi, 2023).

3.1.2. Nonlinear case: the interquantile range The interquartile range that Blundell et al. (2007) report is a difference of two quantiles resulting in moment inequalities that are not affine in the parameter. Fix $0 < \alpha_1 < \alpha_2 < 1$, and suppose that the distribution of W is continuous. The target parameter is the pair of population quantiles,

$$\theta_0 := (F_0^{-1}(\alpha_1), F_0^{-1}(\alpha_2))', \quad F_0(\theta_{0,j}) = \alpha_j, \quad j = 1, 2, \quad (3.10)$$

and the interquantile range is $\theta_{0,2} - \theta_{0,1}$. The parameter space is $\Theta := \{\theta \in [\underline{w}, \bar{w}]^2 : \theta_1 \leq \theta_2\}$ for an interval $[\underline{w}, \bar{w}]$ that contains the support of W .

Write

$$F_0^{\text{lo}}(w) := E[F_0^1(w | X)], \quad F_0^{\text{up}}(w) := E[F_0^1(w | X) p_0(X) + 1 - p_0(X)],$$

so that $\ell_{0,k} = F_0^{\text{lo}}(w_k)$, $u_{0,k} = F_0^{\text{up}}(w_k)$, and $F_0^{\text{lo}}(w) \leq F_0(w) \leq F_0^{\text{up}}(w)$ for every w . Evaluating these bounds at $w = \theta_{0,j}$, and the increment restriction (3.5) at the thresholds $\theta_{0,1} < \theta_{0,2}$, gives the identified set

$$\Theta_I = \left\{ \theta \in \Theta : F_0^{\text{lo}}(\theta_j) \leq \alpha_j \leq F_0^{\text{up}}(\theta_j), \quad j = 1, 2, \quad F_0^{\text{up}}(\theta_2) - F_0^{\text{up}}(\theta_1) \leq \alpha_2 - \alpha_1 \right\}. \quad (3.11)$$

Suppose in addition that $\min\{\alpha_1, \alpha_2 - \alpha_1\} > 1 - P(D = 1)$, F_0^{lo} and F_0^{up} are continuous, and strictly increasing near the points at which they equal α_1 or α_2 . The coordinate projections of Θ_I are then the quantile bounds $[\underline{\theta}_j, \bar{\theta}_j]$, where $F_0^{\text{up}}(\underline{\theta}_j) = \alpha_j$ and $F_0^{\text{lo}}(\bar{\theta}_j) = \alpha_j$.

The increment restriction can make the joint bound on the interquantile range strictly tighter than the difference of the two coordinate bounds. Since

$$F_0^{\text{up}}(w) - F_0^{\text{lo}}(w) = E[(1 - p_0(X))(1 - F_0^1(w | X))],$$

the box corner $(\underline{\theta}_1, \bar{\theta}_2)$, which uniquely maximizes $\theta_2 - \theta_1$ over the Cartesian product of the coordinate bounds, satisfies

$$F_0^{\text{up}}(\bar{\theta}_2) - F_0^{\text{up}}(\underline{\theta}_1) = \alpha_2 - \alpha_1 + E[(1 - p_0(X))(1 - F_0^1(\bar{\theta}_2 | X))].$$

Hence this corner is excluded whenever the last expectation is positive. Since Θ_I is compact, the upper bound on the interquantile range is then strictly smaller than $\bar{\theta}_2 - \theta_1$. This is the unconditional analogue of the cdf coherence argument used by Blundell et al. (2007, Section 2.1.3) to tighten bounds on conditional within-group inequality.

In the notation of Section 2, the observation is $Z_i := (X_i, D_i, D_i W_i)$, and the moment function has five coordinates, obtained by evaluating the scores (3.7) at the parameter,

$$\begin{aligned} g_j^{\text{lo}}(Z; \theta, \xi) &:= \phi_L(Z; \theta_j, \xi) - \alpha_j, & g_j^{\text{up}}(Z; \theta) &:= \alpha_j - \phi_U(Z; \theta_j), & j = 1, 2, \\ g^{\text{inc}}(Z; \theta) &:= D \mathbf{1}\{\theta_1 < W \leq \theta_2\} - (\alpha_2 - \alpha_1). \end{aligned} \quad (3.12)$$

Because

$$\mathbb{E} \phi_L(Z; w, \xi_0) = F_0^{\text{lo}}(w), \quad \mathbb{E} \phi_U(Z; w) = F_0^{\text{up}}(w),$$

and, for $\theta_1 \leq \theta_2$,

$$\mathbb{E}[D \mathbf{1}\{\theta_1 < W \leq \theta_2\}] = F_0^{\text{up}}(\theta_2) - F_0^{\text{up}}(\theta_1),$$

the zero set of the population criterion generated by (3.12) is (3.11). The nuisance parameter is $\xi_0 = (p_0, F_0^1)$. The bias identity in (3.9) holds at every threshold w , so the lower-bound moments are Neyman-orthogonal pointwise in θ ; Assumption 4.5 supplies the uniform control over θ required by the asymptotic theory.

Three features of (3.12) distinguish the interquantile case from the linear case. First, the moments are non-affine in θ , because the quantile locations enter through both the indicator functions and $F^1(\theta_j | X)$. Second, the nuisance is evaluated at the parameter, so $F_0^1(w | x)$ must be estimated uniformly over w . Third, Θ_I need not be convex: the increment restriction defines a nonlinear upper boundary for θ_2 as a function of θ_1 , which need not be concave. Recovering quantiles from the fixed-threshold system of Section 3.1.1 would instead require a threshold grid whose mesh tends to zero, and therefore a growing number of inequalities. Alternatively, one could estimate F_0^{lo} and F_0^{up} uniformly and invert the estimated bound functions. The criterion approach avoids this separate inversion step: the quantile locations enter directly as parameters, and the zero set of the criterion yields Θ_I even when that set is nonconvex.

3.2. Wage gap with an interval-censored outcome

Our second example is the gender wage gap when the wage is recorded only as a bracket, a classic source of partial identification (Manski and Tamer, 2002). We describe the set of coefficients consistent with the brackets by a family of moment inequalities.

The target is the coefficient on the treatment in a partially linear projection of the log wage, of which only a bracket is observed. The projection is motivated by the model

$$Y = D' \theta_0 + f_0(X) + U, \quad \mathbb{E}[U | X, D] = 0, \quad (3.13)$$

where $D \in \mathbb{R}^d$ is a low-dimensional treatment vector, f_0 is an integrable function, and $X \in \mathbb{R}^p$ is a vector of controls whose dimension may be large ($p \gg N$). The target parameter is the partial-linear projection coefficient $\theta_0 := \Sigma_0^{-1} \mathbb{E}[V(\eta_0)Y]$, with $V(\eta_0)$ and Σ_0 as in (3.15). It equals the coefficient in (3.13) when the model holds and remains well defined when effects are heterogeneous. The latent value of Y is bracketed by the observable bounds,

$$Y_L \leq Y \leq Y_U, \quad \Delta := Y_U - Y_L, \quad (3.14)$$

where the width Δ can be random.

The identified set is the set of partial-linear projection coefficients generated by outcomes consistent with the brackets,

$$\Theta_I^* := \{\Sigma_0^{-1}E[V(\eta_0)Y] : Y_L \leq Y \leq Y_U \text{ a.s.}\}, \quad (3.15)$$

where $\eta_0(X) := E[D | X]$, $V(\eta) := D - \eta(X)$ is the residualized treatment and $\Sigma_0 := E[V(\eta_0)V(\eta_0)']$. Let $\ell_0(X) := E[Y_L | X]$ and $U_L := Y_L - \ell_0(X)$ be the residualized lower bracket. By definition, $\Sigma_0\theta_0 = E[V(\eta_0)Y]$, and decomposing $Y = Y_L + R$ with $R := Y - Y_L$ splits this into an observed and an unobserved part,

$$\Sigma_0\theta_0 = E[V(\eta_0)U_L] + E[V(\eta_0)R], \quad (3.16)$$

where the first term uses $E[V(\eta_0) | X] = 0$.

Bounding the unobserved term produces one supporting half-space of Θ_I^* for every direction q . Since $0 \leq R \leq \Delta$, the bound $(q'V(\eta_0))R \leq \Delta(q'V(\eta_0))_+$ holds pointwise for every q , where $a_+ := \max\{0, a\}$, and taking expectations gives

$$q'\Sigma_0\theta_0 \leq E[(q'V(\eta_0))U_L] + E[\Delta(q'V(\eta_0))_+]. \quad (3.17)$$

The set Θ_I^* is convex, so it is the intersection of its supporting half-spaces: (3.17) is the supporting half-space in direction q , and intersecting over all q on the unit sphere recovers Θ_I^* . For implementation we use a finite grid $q_1, \dots, q_L$ and let Θ_I denote the resulting polyhedral outer approximation,

$$\Theta_I = \{\theta \in \Theta : q'_l\Sigma_0\theta \leq E[(q'_lV(\eta_0))U_L] + E[\Delta(q'_lV(\eta_0))_+], \quad l = 1, \dots, L\}. \quad (3.18)$$

Each direction gives one moment inequality, with its own orthogonalization because its nuisance enters through $q'V(\eta)$.

In the notation of Section 2, the observation is $Z_i := (X_i, D_i, Y_{L,i}, Y_{U,i})$, $i = 1, \dots, N$. Consider recentering the positive-part term in (3.17) by $c(X)(q'V(\eta))$ for a measurable function c , which leaves the expectation at η_0 unchanged because $E[V(\eta_0) | X] = 0$. Whenever the positive-part expectation is pathwise differentiable at η_0 , perturbing η in the direction δ gives the derivative at η_0 ,

$$-E[(q'\delta(X))\{\pi_q(\eta_0, X) - c(X)\}], \quad \pi_q(\eta, X) := E[\Delta \mathbf{1}\{q'V(\eta) > 0\} | X], \quad (3.19)$$

which vanishes along every direction δ at $c(X) = \pi_q(\eta_0, X)$. The orthogonal moment function is therefore

$$g_q(Z; \theta, \xi) := (q'V(\eta))(V(\eta)'\theta) - (q'V(\eta))U_L - \left[\Delta(q'V(\eta))_+ - (q'V(\eta))\pi_q(\eta, X)\right]. \quad (3.20)$$

Collecting $g_1, \dots, g_L$ gives a moment function $g(Z; \theta, \xi)$ of length L , and the identified set (2.1) for θ_0 is (3.18). The first-stage nuisance parameter is $\xi_0 := (\eta_0, \ell_0, \pi_{0,1}, \dots, \pi_{0,L})$, estimated by cross-fitting.

4. FORMAL RESULTS

The three corollaries of this section apply Algorithm 2.1 to the moment functions of Section 3 and state, under primitive conditions, the rate of the set estimator and the coverage of the confidence region.

4.1. Wage distribution under selection into employment

Assumption 4.1 collects the restrictions that define the identified set (3.6). Assumption 4.2 is the strict overlap that the inverse propensity in (3.7) requires. Assumption 4.3

places standard restrictions on the convergence rates of the nuisance parameter estimators.

ASSUMPTION 4.1. (IDENTIFICATION) (1) For $j = 1, \dots, S$, the employed-wage distribution stochastically dominates the unemployed-wage distribution, that is, $F_0^1(w_j | X) \leq F_0^0(w_j | X)$ almost surely. (2) The parameter space is $\Theta = [0, 1]^S$; under (1) the polytope (3.6) contains θ_0 and is non-empty.

ASSUMPTION 4.2. (OVERLAP) There exists $\underline{p} > 0$ such that $p_0(X) \geq \underline{p}$ almost surely and every fold estimate satisfies $\hat{p}^{(k)}(X) \geq \underline{p}$ almost surely, $k = 1, \dots, K$.

ASSUMPTION 4.3. (CONVERGENCE RATES) There exist sequences g_N^p and g_N^F such that, with probability approaching one, $\|\hat{p}^{(k)} - p_0\|_{P,2} \leq g_N^p$ and $\max_{j \leq S} \|\hat{F}^{1(k)}(w_j | \cdot) - F_0^1(w_j | \cdot)\|_{P,2} \leq g_N^F$ for every fold k , and

$$g_N^p g_N^F = o(N^{-1/2}), \quad g_N^p \vee g_N^F = o((\log N)^{-1/2}).$$

COROLLARY 4.1. (WAGE DISTRIBUTION UNDER SELECTION) Suppose Assumptions 4.1 to 4.3 hold, and let g collect the $2S$ coordinate moments built from (3.7) and the $S - 1$ increment moments (3.8). Let $\hat{\Theta}_I = C_N(c_N; \hat{\xi})$ be the set estimator of Definition 2.1 at the level $c_N = \log N$. Then the following hold:

- (1) $P(\Theta_I \subseteq \hat{\Theta}_I) \rightarrow 1$ and $d_H(\hat{\Theta}_I, \Theta_I) = O_P(\sqrt{\log N/N})$.
- (2) Let $\tau \in (0, 1)$ be such that the limit law of $\sup_{\theta \in \Theta_I} N Q_N(\theta, \xi_0)$ puts mass less than $1 - \tau$ at zero and has a distribution function that is strictly increasing in a neighbourhood of its $(1 - \tau)$ -quantile, and let $b \rightarrow \infty$ with $b \log N/N \rightarrow 0$. The critical value $\hat{c}_\tau$ of Algorithm 2.1 then satisfies

$$P(\Theta_I \subseteq C_N(\hat{c}_\tau; \hat{\xi})) = 1 - \tau + o(1).$$

- (3) Suppose Θ_I has non-empty interior. Then $d_H(C_N(\hat{c}'; \hat{\xi}), \Theta_I) = O_P(N^{-1/2})$ for every level $\hat{c}' = O_P(1)$ that is at least $N \min_{\theta \in \Theta} Q_N(\theta, \hat{\xi})$ with probability approaching one. The level $\hat{c}' = 0$ is admissible whenever the sample criterion attains zero with probability approaching one.

Corollary 4.1 gives the asymptotic theory for the set estimator and the confidence region in the wage-distribution example. The rate in (1) and the coverage in (2) match the infeasible criterion $Q_N(\theta, \xi_0)$ of Chernozhukov et al. (2007), and (3) is their degenerate case. The first stage converges more slowly than $N^{-1/2}$ but does not affect the first-order asymptotics: Assumption 4.3 restricts only the product of the two first-stage errors, so an $o(N^{-1/4})$ rate for each suffices.

4.2. Interquantile range under selection into employment

Assumption 4.4 extends the stochastic-dominance restriction of Assumption 4.1 to every threshold and requires the upper bound function F_0^{up} of Section 3.1.2 to increase at a positive rate near the quantile bounds. Assumption 4.5 restates Assumption 4.3 uniformly over thresholds. Distribution regression, which estimates $F_0^1(w | x)$ at each threshold w

by an ℓ_1 -penalized logistic regression of $\mathbf{1}\{W \leq w\}$ on X among the employed, attains rates of this type uniformly over thresholds (Belloni et al., 2017). Its fitted values need not be monotone in w . Rearranging them in w restores monotonicity and does not increase $\sup_w |\hat{F}^{1(k)}(w | x) - F_0^1(w | x)|$ at any x (Chernozhukov et al., 2010), so it preserves the rate condition whenever that condition holds in the stronger norm $\|\sup_w |\cdot|\|_{P,2}$.

ASSUMPTION 4.4. (IDENTIFICATION) (1) For every w , the employed-wage distribution stochastically dominates the unemployed-wage distribution, that is, $F_0^1(w | X) \leq F_0^0(w | X)$ almost surely. (2) The distribution of W is continuous, its support lies in $[\underline{w}, \bar{w}]$, and given $D = 1$ and X the wage has a density $f_0^1(w | X)$ bounded by a constant $\bar{f}$. (3) The quantile levels satisfy $\min\{\alpha_1, \alpha_2 - \alpha_1\} > 1 - \mathbb{P}(D = 1)$, and there exist $\bar{\delta} > 0$ and $\underline{f} > 0$ such that $\mathcal{J} := [\underline{\theta}_1 - \bar{\delta}, \bar{\theta}_2 + \bar{\delta}] \subset (\underline{w}, \bar{w})$ and $\mathbb{E}[p_0(X)f_0^1(w | X)] \geq \underline{f}$ for almost every $w \in \mathcal{J}$.

ASSUMPTION 4.5. (CONVERGENCE RATES) For every fold k and every x , the estimate $\hat{F}^{1(k)}(\cdot | x)$ is non-decreasing on $[\underline{w}, \bar{w}]$ with values in $[0, 1]$. There exist sequences g_N^P and g_N^F that satisfy the rate conditions of Assumption 4.3 and such that, with probability approaching one, $\|\hat{p}^{(k)} - p_0\|_{P,2} \leq g_N^P$ and $\sup_{w \in [\underline{w}, \bar{w}]} \|\hat{F}^{1(k)}(w | \cdot) - F_0^1(w | \cdot)\|_{P,2} \leq g_N^F$ for every fold k .

COROLLARY 4.2. (INTERQUANTILE RANGE UNDER SELECTION) Suppose Assumptions 4.2, 4.4 and 4.5 hold, and let g collect the five moments in (3.12). Then conclusions (1), (2) and (3) of Corollary 4.1 hold.

Corollary 4.2 shows that the conclusions of Corollary 4.1 continue to hold for the non-affine moment system (3.12). Because the lower-bound moments evaluate $\hat{F}^1$ at the unknown quantile locations, the first-stage rate must hold uniformly over thresholds. The lower density bound in Assumption 4.4(3), together with the upper density bound in Assumption 4.4(2), provides the local linear error bound that is automatic for the affine system of Corollary 4.1. In particular, the transformation

$$\Phi(\theta) = (F_0^{\text{up}}(\theta_1), F_0^{\text{up}}(\theta_2))$$

is locally bi-Lipschitz on the relevant region, and the transformed identified set is a convex polygon, so Hoffman's bound yields a population moment violation proportional to $d(\theta, \Theta_I)$. Let $r(\theta) := \theta_2 - \theta_1$ and $\mathcal{R}_I := r(\Theta_I)$. Although Θ_I need not be convex, it is nevertheless connected. Specifically, under the transformation Φ used in the proof of Corollary 4.2, it is the inverse image of a convex polygon, and as a result, $\mathcal{R}_I$ is an interval. Since r is Lipschitz, $d_H(r(\hat{\Theta}_I), \mathcal{R}_I) \leq \sqrt{2}d_H(\hat{\Theta}_I, \Theta_I)$, so that the estimated interquantile-range set converges at no slower rate than the rate in Corollary 4.2(1). A confidence interval for the identified interquartile range is obtained from the interval hull

$$\text{CI}_\tau := \left(\inf_{\theta \in C_N(\hat{c}_\tau; \hat{\xi})} r(\theta), \sup_{\theta \in C_N(\hat{c}_\tau; \hat{\xi})} r(\theta) \right).$$

Whenever $C_N(\hat{c}_\tau; \hat{\xi})$ contains Θ_I , CI_τ contains $\mathcal{R}_I$. Therefore, $\liminf_{N \rightarrow \infty} \mathbb{P}(\mathcal{R}_I \subseteq \text{CI}_\tau) \geq 1 - \tau$. This projection is generally conservative for the scalar target because $\hat{c}_\tau$ is calibrated to cover the entire set Θ_I . By contrast, Kaido et al. (2019) calibrate inference directly for a component or smooth function of θ .

4.3. Wage gap with an interval-censored outcome

Assumption 4.6 imposes a standard identification condition and ensures that the finite system of directional inequalities defines a bounded polyhedron. Assumption 4.7 provides bounded envelopes for the moments. The only condition specific to the non-smooth positive-part term is Assumption 4.8, which requires its population remainder to be quadratic in the propensity error. Finally, Assumption 4.9 requires the resulting first-stage bias to be $o(N^{-1/2})$ and the L^2 nuisance errors to satisfy the continuity rate.

ASSUMPTION 4.6. (IDENTIFICATION) (1) The matrix $\Sigma_0 := E[(D - \eta_0(X))(D - \eta_0(X))']$ is non-singular. (2) The finite set $q_1, \dots, q_L \in \mathbb{S}^{d-1}$ is symmetric about the origin, and spans $\mathbb{R}^d$. (3) Θ is compact and contains the polyhedron cut out by the L half-spaces (3.17).

ASSUMPTION 4.7. (BOUNDEDNESS) For finite constants $\bar{D}, \bar{Y}, \bar{\Delta}$ it holds that $\|D\| \leq \bar{D}$, $|Y_L| \leq \bar{Y}$ and $\Delta \leq \bar{\Delta}$ almost surely.

ASSUMPTION 4.8. (QUADRATIC KINK REMAINDER) For every $l = 1, \dots, L$ there exists $C_K < \infty$ such that for every η with $\|\eta(X)\| \leq \bar{D}$ almost surely,

$$|\mathcal{K}_l(\eta)| \leq C_K \|\eta - \eta_0\|_{P,2}^2, \quad (4.1)$$

where

$$\mathcal{K}_l(\eta) := E[\Delta((q'_l V(\eta))_+ - (q'_l V(\eta_0))_+ + q'_l(\eta - \eta_0)(X) \mathbf{1}\{q'_l V(\eta_0) > 0\})].$$

ASSUMPTION 4.9. (CONVERGENCE RATES) Every fold estimate satisfies $\|\hat{\eta}^{(k)}(X)\| \leq \bar{D}$, $|\hat{\ell}^{(k)}(X)| \leq \bar{Y}$ and $0 \leq \hat{\pi}_l^{(k)}(X) \leq \bar{\Delta}$ almost surely, $l = 1, \dots, L$. There exist sequences $g_{\eta,N}, g_{\ell,N}, g_{\pi,N}$ such that with probability approaching one for every fold k , $\|\hat{\eta}^{(k)} - \eta_0\|_{P,2} \leq g_{\eta,N}$, $\|\hat{\ell}^{(k)} - \ell_0\|_{P,2} \leq g_{\ell,N}$, $\max_{l \leq L} \|\hat{\pi}_l^{(k)} - \pi_{0,l}\|_{P,2} \leq g_{\pi,N}$, and

$$g_{\eta,N}(g_{\eta,N} \vee g_{\ell,N} \vee g_{\pi,N}) = o(N^{-1/2}), \quad g_{\eta,N} \vee g_{\ell,N} \vee g_{\pi,N} = o((\log N)^{-1/2}).$$

COROLLARY 4.3. (WAGE GAP WITH AN INTERVAL-CENSORED WAGE) Suppose Assumptions 4.6 to 4.9 hold, and let $g = (g_1, \dots, g_L)$ be built from (3.20). Then conclusions (1), (2) and (3) of Corollary 4.1 hold.

The role of Assumption 4.8 is to control observations near the kink $q'_l V(\eta_0) = 0$. It can be verified in different ways depending on the design. For continuous designs, a conventional margin or anti-concentration condition around zero implies the quadratic bound. In discrete designs, the same condition may instead hold through sign stability: if perturbing η does not change the sign of $q'_l V(\eta)$, then the kink remainder is exactly zero. The empirical application of Section 5 is a particularly simple special case. Conditional on X , D has two support points, and the estimated propensity remains in $(0, 1)$. Perturbing the propensity therefore moves the two residual support points without changing their signs, so the kink remainder vanishes exactly. Moreover, with constant bracket width Δ , each π_l is available as a closed-form function of the scalar propensity. Consequently, the general rate condition stated in Assumption 4.9 reduces to $g_{\eta,N}(g_{\eta,N} + g_{\ell,N}) = o(N^{-1/2})$.

5. EMPIRICAL APPLICATION

We use the interval-augmented March Supplement of the 2015 Current Population Survey data where we intentionally bracket an observed outcome (log wage) to demonstrate our procedure. The sample is restricted to white non-Hispanic individuals aged 25–64 who worked more than 35 hours per week for at least 50 weeks, resulting in a total of $N = 32,523$ observations. The outcome Y is the log hourly wage, and the treatment vector is $D = (\text{female}, \text{female} \times \text{college})'$. Therefore, θ_1 measures the gender wage gap among non-college workers, and θ_2 measures the additional gender gap differential associated with college. The conditioning vector X contains 259 demographic, region, and experience controls and their interactions. We construct the interval-censored wage while retaining the observed wage as an infeasible benchmark. For each bracket width $\Delta \in \{1, 2, 3\}$, we replace Y by lower and upper endpoints Y_L and Y_U satisfying $\Delta = Y_U - Y_L$. Thus the information available to the estimator is only that $Y \in [Y_L, Y_U]$, while observed Y is retained for the subsequent analysis.

For each Δ , we apply the procedure of Sections 3.2 and 2.2. The first-stage nuisance functions are the propensity $\eta_{0,1}(X) = P(\text{female} = 1 \mid X)$, and the conditional mean of the lower bracket, $\ell_0(X) = E[Y_L \mid X]$. Both nuisance components are estimated by ℓ_1 -penalized regressions with $K = 2$ fold cross-fitting and the penalty selection rule of Belloni et al. (2017). Note that the second component of the treatment propensity satisfies $\eta_{0,2}(X) = \eta_{0,1}(X) \cdot \text{college}$. We impose the orthogonal moment inequality (3.20) at $L = 180$ equally spaced directions on the unit circle. For inference, we form the preliminary contour at $c_N = \log N$ and calibrate the final level by the balanced-block subsampling procedure of Definition 2.2. We set $b = 160$, giving $B_N = 203$ disjoint blocks, with each block drawing the same number of observations from each cross-fitting fold. The reported confidence region uses the 0.95 empirical quantile of the block statistics.

We contrast our method with two other procedures: the estimated set based on the non-orthogonal moment function and the infeasible regression using the observed wage. Table 1 reports the results along the estimated identified set based on the orthogonal moment function and its 95% confidence region. The comparison between the orthogonal and non-orthogonal set estimators reveals only a marginal difference. In this application, the numerical difference is small with at most 0.003Δ difference in each coordinate projection across all bracket widths. The negligible difference can be explained by the sufficiently accurate estimates of the propensity scores in the first step. Finally, a point-estimate resulting from the infeasible regression is well-covered by the estimated sets.

Figure 1 shows the estimated sets for the three bracket widths. First, it reflects how loss of information created by interval censoring influences the size of the estimated sets. Larger brackets necessarily result in larger estimated sets. Second, it describes how beneficial joint set estimation is compared to its coordinatewise projections imposed independently. Across all three values of Δ , the area of the estimated set $\hat{\Theta}_I$ is approximately one half of the area of the set resulting from the coordinatewise independent bounds. This discrepancy is a result of dependence between the two coefficient restrictions that is discarded when restrictions are imposed independently.

Two qualifications are important. First, the directional inequalities characterize the outer identified set associated with the restrictions developed in Section 3.2; as discussed in Semenova (2023), this set need not be sharp for the underlying structural coefficient θ_0 . Second, implementation replaces the continuum of directions by $L = 180$ directions,

so the reported estimated set is itself a finite-direction outer approximation. Increasing the number of directions might therefore produce better approximations of the outer set.

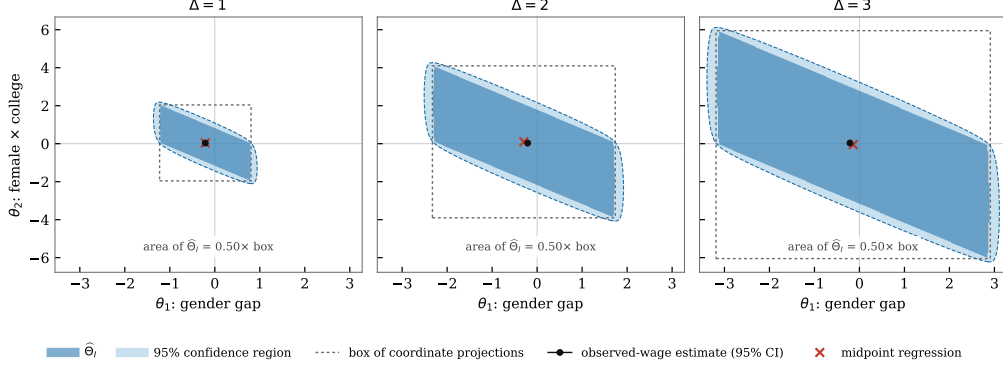


Figure 1: Estimated identified set and 95% confidence region for the bracket widths $\Delta \in \{1, 2, 3\}$.

Table 1: Bounds on the gender wage gap and its college interaction with a bracketed log wage

Δ		θ_1 (gender gap)	θ_2 (female $\times$ college)	area / box	$\widehat{c}_\tau$
1	$\widehat{\Theta}_I$	$[-1.227, 0.805]$	$[-1.960, 2.040]$	0.496	16.61
	95% region	$[-1.363, 0.941]$	$[-2.103, 2.184]$		
	uncorrected moment	$[-1.229, 0.807]$	$[-1.957, 2.037]$		
2	$\widehat{\Theta}_I$	$[-2.327, 1.736]$	$[-3.903, 4.097]$	0.496	23.41
	95% region	$[-2.511, 1.920]$	$[-4.073, 4.268]$		
	uncorrected moment	$[-2.332, 1.741]$	$[-3.896, 4.091]$		
3	$\widehat{\Theta}_I$	$[-3.187, 2.909]$	$[-6.050, 5.950]$	0.496	24.31
	95% region	$[-3.390, 3.112]$	$[-6.224, 6.124]$		
	uncorrected moment	$[-3.194, 2.916]$	$[-6.041, 5.941]$		
observed-wage estimate		-0.210 (0.008)	0.033 (0.015)		

Notes. CPS 2015, $N = 32,523$ workers, $p = 259$ controls. Rows labelled $\hat{\Theta}_I$ and 95% region report the coordinate projections of the estimated set and of the confidence region $C_N(\hat{c}_\tau; \hat{\xi})$ with $\tau = 0.05$. The column area / box is the ratio of the area of $\hat{\Theta}_I$ to the area of that box. The critical value $\hat{c}_\tau$ is the $(1 - \tau)$ -quantile of $B_N = 203$ disjoint block statistics of size $b = 160$, balanced across the two folds. The row “uncorrected moment” reports the projections obtained from the non-orthogonal specification. The observed-wage estimate is infeasible and is reported for reference, with heteroskedasticity-robust standard errors in parentheses.

6. CONCLUSION

This paper develops estimation and inference for partially identified models in which the moment inequalities defining the identified set depend on point-identified nuisance

functions estimated in a first stage. The approach combines Neyman-orthogonal moment inequalities, cross-fitting, and the criterion-function representation of partially identified sets. Under the stated conditions, estimation of the nuisance functions is second order and the feasible contour set inherits the first-order Hausdorff rate of its infeasible counterpart. For inference, we develop a disjoint-block subsampling procedure adapted to the cross-fitted first stage. It yields asymptotically valid pointwise coverage without strengthening the first-stage small-bias requirement used for set estimation. The examples illustrate different aspects of the framework. In the wage-selection application, integrating the conditional restrictions underlying Blundell et al. (2007) yields unconditional moment inequalities that can accommodate a rich, machine-learned conditioning set. The interquantile-range problem shows that the same framework extends beyond affine systems. The resulting identified set need not be convex, yet it can be estimated directly as the zero set of the criterion and projected onto the interquantile range without separately inverting estimated bound functions. Finally, the interval-outcome example illustrates a different source of nonregularity. The general theory proposed herein allows continuous, discrete, or mixed treatment designs provided the population remainder associated with the positive-part kink is quadratic in the first-stage error.

The results also indicate several limits of the present analysis. The subsampling guarantees are pointwise in the data-generating process, and the number of moment inequalities is fixed in the asymptotic theory. Developing uniformly valid calibration with regularized first stages, allowing the number of thresholds or directions to increase with the sample size, and extending the theory to moment systems with more general forms of parameter-dependent non-smoothness are natural directions for further work.

Appendix A and Appendix B are intended for online publication only.

APPENDIX A: ASYMPTOTIC THEORY

A.1. Conditions on the moment function and the first stage

We take the moment function to be Neyman-orthogonal in the sense of Definition 2.3, on a set of nuisance values that contains the first-stage estimates with probability approaching one. The precise requirement is part (d) of Assumption A.3 below. The paper posits an orthogonal moment rather than constructing one. For point-identified parameters the construction is largely automatic. Orthogonal moments for many functionals have been derived, or can be derived, following Chernozhukov et al. (2018) and Chernozhukov et al. (2022). A smooth functional of a nonparametric regression has a Riesz representer, and Chernozhukov et al. (2022b,a) estimate that representer directly from the data, so the orthogonal moment is produced by the procedure rather than derived by hand for each application.

Assumption A.1 ensures that the population criterion distinguishes the boundary of the identified set: once θ is bounded away from Θ_I , the moment $Eg(Z; \theta, \xi_0)$ is bounded away from zero by an amount proportional to $d(\theta, \Theta_I)$. It is the population analogue of Condition C.2 of Chernozhukov et al. (2007, p. 1253), the existence of a polynomial minorant, which bounds the sample criterion below by $\kappa[d(\theta, \Theta_I) \wedge \delta]^\gamma$ outside an $a_n^{-1/\gamma}$ -neighbourhood of Θ_I . Its counterpart in Chernozhukov et al. (2013) is Condition V, $\theta_n(v) - \theta_{n0} \geq (c_n d(v, V_0))^{\rho_n} \wedge \delta$, stated there for an index set V_0 rather than for a parameter. We depart from these conditions in two respects. First, we impose the bound on the population moment rather than on the sample criterion. Second, we fix the exponent at one on the moment, hence $\gamma = 2$ on the criterion, and Appendix B.4 shows that this is not a restriction for affine moment systems or for the interquantile moments of Section 3.1.2. We keep the truncation at $\delta_{\min}$, as both of them do, so nothing is required of θ far from Θ_I beyond a fixed positive floor.

ASSUMPTION A.1. (IDENTIFIABILITY) *The identified set Θ_I is non-empty, and there exist constants $C_{\min} > 0$ and $\delta_{\min} > 0$ such that*

$$\|Eg(Z; \theta, \xi_0)\|_+ \geq C_{\min} [d(\theta, \Theta_I) \wedge \delta_{\min}], \quad \forall \theta \in \Theta. \quad (\text{A.1})$$

Assumption A.2 is a functional central limit theorem for the moment function evaluated at the true nuisance.

ASSUMPTION A.2. (DONSKEER PROPERTY) *In the space $\ell^\infty(\Theta)^L$,*

$$\mathbb{G}_N g(Z_i; \theta, \xi_0) := \sqrt{N}(E_N g(Z_i; \theta, \xi_0) - E g(Z_i; \theta, \xi_0)) \Rightarrow \mathbb{G}(\theta), \quad (\text{A.2})$$

where $\mathbb{G}(\theta)$ is a mean-zero Gaussian process on Θ with almost surely continuous paths.

Assumption A.3 controls the impact of first-stage estimation. It is stated on a sequence of neighbourhoods $\Xi_N \subset \Xi$ of ξ_0 that contain the estimate $\hat{\xi}$ with probability approaching one and shrink as N grows. The rate of shrinkage g_N measures the quality of the first stage. The assumption combines Neyman orthogonality, which removes the first-order effect of the estimation error $\hat{\xi} - \xi_0$, with a bound on the second-order remainder, condition (e).

ASSUMPTION A.3. (ORTHOGONALITY AND FIRST-STAGE QUALITY) *There exists a sequence of neighbourhoods $\Xi_N \subset \Xi$ of ξ_0 such that the following hold.*

- (a) (Truth in the neighbourhood.) *The true value ξ_0 belongs to Ξ_N for all $N \geq 1$.*
- (b) (Estimator in the neighbourhood.) *There exists a sequence $\phi_N = o(1)$ such that every fold estimate $\widehat{\xi}^{(k)}$, $k = 1, \dots, K$, belongs to Ξ_N with probability at least $1 - \phi_N$. We write $\mathcal{B}_N := \{\widehat{\xi}^{(k)} \in \Xi_N \text{ for all } k\}$, so that $P(\mathcal{B}_N) \geq 1 - K\phi_N \rightarrow 1$.*
- (c) (Consistency.) *The neighbourhoods shrink:*

$$g_N := \sup_{\xi \in \Xi_N} \|\xi - \xi_0\|_{P,2} = o(1).$$

- (d) (Orthogonality.) *The moment function $g(Z; \theta, \xi)$ is Neyman-orthogonal at ξ_0 on Ξ_N in the sense of Definition 2.3, for every $\theta \in \Theta$.*
- (e) (Small bias.) *There exists a sequence $\bar{s}_N = o(N^{-1/2})$ bounding the bias induced by the first stage:*

$$\sup_{\theta \in \Theta} \sup_{\xi \in \Xi_N} \|E g(Z; \theta, \xi) - E g(Z; \theta, \xi_0)\| \leq \bar{s}_N.$$

Only conditions (b) and (e) enter the proofs. Condition (e) is the only channel through which the first stage enters the bias. Conditions (a), (c) and (d) describe the mechanism that makes it attainable at the rate $o(N^{-1/2})$, and they are verified alongside it in Appendix B.4. When $r \mapsto E g(Z; \theta, \xi_0 + r(\xi - \xi_0))$ is twice differentiable, condition (d) removes the first-order term and a Taylor expansion gives

$$\|E g(Z; \theta, \xi) - E g(Z; \theta, \xi_0)\| \leq \frac{1}{2} \sup_{r \in [0,1]} \|\partial_r^2 E g(Z; \theta, \xi_0 + r(\xi - \xi_0))\|, \quad (\text{A.3})$$

so a bound on the second Gateaux derivative implies condition (e). When that derivative is a bounded bilinear form in the nuisance error, condition (e) holds with $\bar{s}_N = Cg_N^2$, which is the product-of-rates condition of the double/debiased machine-learning literature. We state condition (e) as a bias bound rather than a derivative bound because the bracketed-wage moment of Section 3.2 has a kink and is verified in Appendix B.4.5 without differentiability. No rate is imposed in condition (c): the first-stage rate enters the variance only through Assumption A.4(c).

Assumption A.4 requires that the moment function $g(Z; \theta, \xi)$ be regular in θ for each fixed $\xi \in \Xi_N \cup \{\xi_0\}$, with constants that do not depend on ξ . We consider the class $\mathcal{F}_\xi := \{g(\cdot; \theta, \xi) : \theta \in \Theta\}$ and require its uniform covering entropy to be bounded. Uniformity over ξ lets the proofs condition on a fold's estimate and integrate it out.

ASSUMPTION A.4. (REGULARITY OF MOMENT FUNCTION) *The following hold for the class $\mathcal{F}_\xi$, for every N and every $\xi \in \Xi_N \cup \{\xi_0\}$, with constants (c, c_1, a, v) that depend on neither ξ nor N .*

- (a) (Envelope.) *There exists a measurable envelope $F_\xi = F_\xi(Z)$ that bounds all elements in the function class almost surely:*

$$\sup_{\theta \in \Theta} |g_l(Z; \theta, \xi)| \leq F_\xi(Z) \quad \text{a.s.}, \quad l \in \{1, \dots, L\},$$

where $g = (g_1, \dots, g_L)$. Moreover, the envelope F_ξ has a finite c -norm for some $c > 2$: $\|F_\xi\|_{P,c} := (\int |F_\xi(z)|^c dP(z))^{1/c} \leq c_1$.

- (b) (Entropy.) *There exist finite constants $a \geq e$ and $v \geq 1$ such that the uniform covering entropy of the class $\mathcal{F}_\xi$ is bounded:*

$$\sup_{\tilde{Q}} \log \mathcal{N}(\epsilon \|F_\xi\|_{\tilde{Q},2}, \mathcal{F}_\xi, \|\cdot\|_{\tilde{Q},2}) \leq v \log(a/\epsilon), \quad 0 < \epsilon \leq 1, \quad (\text{A.4})$$

where $\mathcal{N}(\epsilon, \mathcal{F}, \|\cdot\|)$ is the ϵ -covering number of $\mathcal{F}$ in the norm $\|\cdot\|$ and the supremum is over finitely discrete probability measures $\tilde{Q}$.

- (c) (Continuity in the nuisance.) *There exists a sequence r'_N obeying $r'_N \sqrt{\log N} = o(1)$ that bounds the mean-square distance between the moment at a nuisance in the neighbourhood and at the truth:*

$$\sup_{\theta \in \Theta} \sup_{\xi \in \Xi_N} (\mathbb{E} \|g(Z; \theta, \xi) - g(Z; \theta, \xi_0)\|^2)^{1/2} \leq r'_N.$$

Assumption A.4 generalizes the regularity assumption in the point-identified moment problem of Chernozhukov et al. (2018) to the partially identified case.

A.2. Rate of convergence

The first result is the rate at which the set estimator of Definition 2.1 converges to Θ_I in Hausdorff distance. Because g is Neyman-orthogonal in ξ , the first-stage estimation error does not affect the first-order asymptotics, and the criterion-function theory of Chernozhukov et al. (2007) applies to the feasible criterion. The rate is that of the infeasible contour set built on the true nuisance, the case $a_n = N, \gamma = 2$ of Theorem 3.1(2) in Chernozhukov et al. (2007). The level must satisfy the two hypotheses of that theorem: it contains the infeasible statistic with probability approaching one, which makes $\hat{\Theta}_I$ cover Θ_I , and it is of smaller order than N .

THEOREM A.1. (CONSISTENCY AND RATE OF CONVERGENCE) *Suppose Assumptions A.1 to A.4 hold. Let $\hat{\Theta}_I$ be the contour set estimator of Definition 2.1 with $\hat{c}$ satisfying*

$$\mathbb{P}\left(\sup_{\theta \in \Theta_I} N Q_N(\theta, \hat{\xi}) \leq \hat{c}\right) = 1 - o(1), \quad \hat{c}/N \rightarrow_p 0. \quad (\text{A.5})$$

Then $\Theta_I \subseteq \hat{\Theta}_I$ with probability approaching one, and the Hausdorff distance between $\hat{\Theta}_I$ and Θ_I satisfies

$$d_H(\hat{\Theta}_I, \Theta_I) = O_P\left(\sqrt{(1 \vee \hat{c})/N}\right). \quad (\text{A.6})$$

Any deterministic level $c_N \rightarrow \infty$ with $c_N/N \rightarrow 0$ satisfies (A.5), because $\sup_{\theta \in \Theta_I} N Q_N(\theta, \hat{\xi}) = O_P(1)$ under the same assumptions (Step 4 of the proof of Lemma B.1).

Theorem A.1 says that the level should be as small as the coverage requirement allows. A level $\hat{c} = O_P(1)$ that covers delivers the parametric rate $N^{-1/2}$, but no feasible level does both in general: the infeasible statistic $\sup_{\theta \in \Theta_I} N Q_N(\theta, \hat{\xi})$ is $O_P(1)$ but unknown, a deterministic level $c_N \rightarrow \infty$ covers but is unbounded, and the subsampling level $\hat{c}_\tau$ of Appendix A.3 is bounded but covers only with probability $1 - \tau$. For $c_N = \log N$ the rate is $\sqrt{\log N/N}$, slower than the parametric rate by the factor $\sqrt{\log N}$.

For the confidence region $C_N(\widehat{c}_\tau; \widehat{\xi})$ the proof of Theorem A.1 still gives the outer bound $\sup_{\theta \in C_N(\widehat{c}_\tau; \widehat{\xi})} d(\theta, \Theta_I) = O_P(N^{-1/2})$, since that direction does not use containment.

The parametric rate is attainable without a covering level when the criterion is *degenerate*: the sample criterion vanishes on data-dependent sets that approximate Θ_I at rate $N^{-1/2}$, so that any bounded level that is at least the sample minimum already captures those sets.

DEFINITION A.1. (DEGENERACY) *There exists a sequence of possibly data-dependent subsets $\Theta_N \subset \Theta$ such that the cross-fitted criterion is minimal on Θ_N with probability approaching one and Θ_N approximates Θ_I at the parametric rate: for all $p \in (0, 1)$, there exists N_p such that for all $N \geq N_p$,*

$$\mathbb{P} \left(\sup_{\theta \in \Theta_N} \{Q_N(\theta, \widehat{\xi}) - \inf_{\tilde{\theta} \in \Theta} Q_N(\tilde{\theta}, \widehat{\xi})\} = 0 \right) \geq 1 - p, \quad d_H(\Theta_N, \Theta_I) = O_P(N^{-1/2}). \quad (\text{A.7})$$

This is Condition C.3 of Chernozhukov et al. (2007) with $a_n = N$ and $\gamma = 2$.

LEMMA A.1. (SUFFICIENT CONDITIONS FOR DEGENERACY) *Suppose Assumptions A.1 to A.4 hold. If in addition Θ_I has non-empty interior and there exist positive constants C, M, δ such that, for all $\epsilon \in [0, \delta]$, $\Theta_I^{-\epsilon} \neq \emptyset$ and*

$$\max_l \mathbb{E} g_l(Z; \theta, \xi_0) \leq -C(\epsilon \wedge \delta) \quad \text{for all } \theta \in \Theta_I^{-\epsilon}, \quad d_H(\Theta_I^{-\epsilon}, \Theta_I) \leq M\epsilon, \quad (\text{A.8})$$

where $\Theta_I^{-\epsilon} := \{\theta \in \Theta_I : B(\theta, \epsilon) \subseteq \Theta_I\}$, with $B(\theta, \epsilon)$ the closed ball in $\mathbb{R}^{d_\theta}$, is the ϵ -contraction of Θ_I in the sense of Chernozhukov et al. (2007), then the degeneracy condition of Definition A.1 holds. Moreover, for any level $\widehat{c}' = O_P(1)$ with

$$\widehat{c}' \geq N \min_{\theta \in \Theta} Q_N(\theta, \widehat{\xi}) \quad \text{with probability approaching one,} \quad (\text{A.9})$$

the contour set satisfies $d_H(C_N(\widehat{c}'; \widehat{\xi}), \Theta_I) = O_P(N^{-1/2})$.

Chernozhukov et al. (2007, p. 1255) impose Condition C.3 rather than derive it, and motivate it by the interval-data examples, where the criterion is identically zero on the interior of the identified set with probability approaching one. Their Theorem 3.2 shows that under it the contour set at any level $\widehat{c}' \geq \inf_{\theta} a_n Q_n(\theta)$ with $\widehat{c}' = O_P(1)$ attains the exact rate $a_n^{-1/\gamma}$. Lemma A.1 is the two-stage version of that result: it gives primitive sufficient conditions for degeneracy when the criterion carries an estimated nuisance, and its level condition (A.9) is the analogue of theirs. The gain is in the *estimation* rate only: a level $\widehat{c}'$ satisfying (A.9) may be far too small to satisfy (A.5), so $C_N(\widehat{c}'; \widehat{\xi})$ is a set estimator and not a confidence region. The level $\widehat{c}' = 0$ is admissible whenever the sample criterion attains zero, which is the interval-data case.

A.3. Subsampling inference

To construct a confidence region $C_N(\widehat{c}_\tau; \widehat{\xi})$ with the coverage property

$$\mathbb{P}(\Theta_I \subseteq C_N(\widehat{c}_\tau; \widehat{\xi})) = 1 - \tau + o(1), \quad (\text{A.10})$$

we must find the asymptotic distribution of the inferential statistic

$$\mathcal{C}_N := \sup_{\theta \in \Theta_I} N Q_N(\theta, \xi_0) \quad (\text{A.11})$$

and use it to estimate the $(1 - \tau)$ -quantile. Define the limiting random variable

$$\mathcal{T} := \sup_{\theta \in \Theta_I} \|I(\theta) + \mathbb{G}(\theta)\|_+^2, \quad (\text{A.12})$$

where $I(\theta)$ is an L -vector with components, and the convention $-\infty + x = -\infty$, $\max\{-\infty, 0\} = 0$ is used inside $\|\cdot\|_+$,

$$I_l(\theta) := \begin{cases} -\infty, & \text{if } \mathbb{E} g_l(Z; \theta, \xi_0) < 0, \\ 0, & \text{if } \mathbb{E} g_l(Z; \theta, \xi_0) = 0. \end{cases} \quad (\text{A.13})$$

The convergence $\mathcal{C}_N \Rightarrow \mathcal{T}$ is Condition C.4 of Chernozhukov et al. (2007), who assume it in their general theory and verify it for moment-condition models in their Section 4, where the limit takes the form (A.12). We follow the same division of labour and state it as Assumption A.6, with continuity of the cdf required on $(0, \infty)$ only, so that an atom at zero is allowed. Theorem A.2 asks in addition that the cdf be strictly increasing near the quantile being estimated, the condition under which the subsampling quantile converges.

The critical value is the subsampling quantile of Definition 2.2, computed at a deterministic preliminary level $c_N \rightarrow \infty$ with $c_N/N \rightarrow 0$, for instance $c_N = \log N$, which delivers the rate $\sqrt{\log N/N}$ of Theorem A.1. We depart from Chernozhukov et al. (2007) in one implementation detail: their Theorem 3.3 is stated for all $\binom{N}{b}$ subsets of size b , with a randomly chosen subcollection allowed in practice, whereas we use $\lfloor N/b \rfloor$ disjoint blocks. The reason is computational: with a cross-fitted first stage, each subsample statistic requires re-evaluating the criterion, so the number of subsamples is the binding cost, and the price is a larger Monte Carlo error in $\hat{G}$, of order $B_N^{-1/2}$. Disjoint blocks are also independent, which Step 2 of the proof of Theorem A.2 uses. The critical value must be an upper quantile: coverage obtains on the event $\{\mathcal{C}_N \leq \hat{c}_\tau\}$, so $\mathbb{P}(\mathcal{C}_N \leq \hat{c}_\tau) \rightarrow 1 - \tau$ requires $\hat{c}_\tau \rightarrow_p q_{1-\tau}(\mathcal{T})$.

Subsampling also requires that the feasible criterion stay close to the infeasible criterion, at the block scale b , on the ϵ_N -expansion of the identified set, $\epsilon_N := \sqrt{(1 \vee c_N)/N}$, where the block statistics are evaluated. The proof needs one condition for this beyond those of Theorem A.1, a linear majorant that reverses Assumption A.1.

ASSUMPTION A.5. (LINEAR MAJORANT) *There exists $C_{\max} < \infty$ such that $\|\mathbb{E} g(Z; \theta, \xi_0)\|_+ \leq C_{\max} d(\theta, \Theta_I)$ for all $\theta \in \Theta$.*

No strengthening of the first-stage conditions is required for coverage. On a block of size b the first-stage bias enters at scale $\sqrt{b} \bar{s}_N$ and the expansion of the identified set at scale $\sqrt{b} \epsilon_N = \sqrt{b(1 \vee c_N)/N}$, the first vanishes because $b \leq N$, and the block-size condition $b c_N/N \rightarrow 0$ of Theorem A.2 makes the second vanish. The full-sample analogue of this step would require $\bar{s}_N = o(N^{-1/2}(\log N)^{-1/2})$ at $c_N = \log N$, which the subsampling construction does not need (Lemma B.3).

Finally, we impose two conditions as Conditions C.4 and C.5 of Chernozhukov et al. (2007), stated at the block size.

ASSUMPTION A.6. (LIMIT DISTRIBUTION) *The infeasible inferential statistic $\mathcal{C}_N :=$*

$\sup_{\theta \in \Theta_I} N Q_N(\theta, \xi_0)$ converges in distribution to the law $\mathcal{T}$ of (A.12), and the cdf of $\mathcal{T}$ is continuous on $(0, \infty)$. An atom at zero is allowed. It occurs exactly when, with positive probability, every binding coordinate of $\mathbb{G}(\theta)$ is non-positive at every $\theta \in \Theta_I$. This is Condition C.4 of Chernozhukov et al. (2007), who verify it for moment-condition models in their Section 4, with continuity required on $(0, \infty)$ rather than on $[0, \infty)$.

ASSUMPTION A.7. (APPROXIMABILITY) Let $\epsilon_N := \sqrt{(1 \vee c_N)/N}$ be the Hausdorff rate of Theorem A.1 at the level c_N , and let $b = b_N \rightarrow \infty$ with $b c_N/N \rightarrow 0$. For every fixed $C_0 > 0$ and every sequence of non-empty compact sets $K_N \subset \Theta$ with $d_H(K_N, \Theta_I) \leq C_0 \epsilon_N$, which may depend on the data of the block, the block statistic $T'_b := \sup_{\theta \in K_N} b Q_{j,b}(\theta, \xi_0)$ computed on a block of size b satisfies

$$P(T'_b \leq x) - P(\mathcal{T} \leq x) = o(1) \quad \text{at every continuity point } x \text{ of the cdf of } \mathcal{T}.$$

The assumption has two parts. The first is that the block statistic built on the true identified set has the limit law $\mathcal{T}$. That is Assumption A.6 applied at sample size b , and it requires only $b \rightarrow \infty$. The second is that replacing Θ_I by a set within $C_0 \epsilon_N$ of it, including the data-dependent sets that the two-sided bounds in the proof of Theorem A.2 produce, does not move the limit law. Here the block size and the Hausdorff rate interact: the perturbation is negligible when $\sqrt{b} \epsilon_N \rightarrow 0$, which for $\epsilon_N \asymp \sqrt{\log N/N}$ holds whenever $b = o(N/\log N)$. It depends on the local geometry of Θ_I near the binding moments; Lemma B.5 gives sufficient conditions.

When $\theta \mapsto g(z; \theta, \xi)$ is discontinuous, as for the interquantile moments (3.12), contour sets need not be compact, and we use a uniform form of Assumption A.7. For $C_0 > 0$, let $\mathcal{S}_N(C_0)$ be the collection of non-empty sets $S \subset \Theta$ with $d_H(S, \Theta_I) \leq C_0 \epsilon_N$, and let $T_b := \sup_{\theta \in \Theta_I} b Q_{j,b}(\theta, \xi_0)$. The condition is that $T_b \Rightarrow \mathcal{T}$ and that, for every fixed $C_0 > 0$, there is a function Δ_b of the block data, the same for every block, with $\Delta_b \rightarrow_p 0$ and

$$\sup_{S \in \mathcal{S}_N(C_0)} \left| \sup_{\theta \in S} \sqrt{b Q_{j,b}(\theta, \xi_0)} - \sqrt{T_b} \right| \leq \Delta_b. \quad (\text{A.14})$$

This condition implies Assumption A.7, since every admissible K_N belongs to $\mathcal{S}_N(C_0)$.

THEOREM A.2. (VALIDITY OF SUBSAMPLING) Suppose Assumptions A.1 to A.4 and Assumptions A.5 and A.6 hold, that either $\theta \mapsto g(z; \theta, \xi)$ is continuous for every z and every $\xi \in \Xi_N \cup \{\xi_0\}$ and Assumption A.7 holds, or (A.14) holds, and that the block size satisfies $b \rightarrow \infty$ with $b c_N/N \rightarrow 0$. Let $\tau \in (0, 1)$, with $1 - \tau$ the nominal confidence coefficient, and suppose $P(\mathcal{T} = 0) < 1 - \tau$ and that the cdf of $\mathcal{T}$ is strictly increasing on a neighbourhood of $q_{1-\tau}(\mathcal{T})$. Then:

- (1) The critical value $\hat{c}_\tau$ of Definition 2.2 converges in probability to the $(1 - \tau)$ -quantile $q_{1-\tau}(\mathcal{T})$ of $\mathcal{T}$, where $\mathcal{T}$ is given in (A.12).
- (2) $P(\Theta_I \subseteq C_N(\hat{c}_\tau; \hat{\xi})) = 1 - \tau + o(1)$.

Theorem A.2 is pointwise in P , where the coverage probability converges to $1 - \tau$ for each fixed P_0 satisfying the assumptions. Subsampling can deliver uniformly valid inference in moment-inequality models, including coverage of the identified set, under conditions given by Romano and Shaikh (2008, 2010) and Andrews and Guggenberger

(2009); bootstrap and simulation alternatives with uniform coverage of the identified set are developed by Bugni (2010) and Chen et al. (2018). Two parts of our argument are stated for fixed P_0 and would have to be strengthened for a uniform result. First, Assumption A.1 holds with constants $C_{\min}$ and $\delta_{\min}$ that are fixed under P_0 ; along drifting sequences P_N in which a moment becomes binding at rate $N^{-1/2}$ these constants degenerate, and the separation used in Step 3 of the proof of Lemma B.1 fails. Second, the first-stage conditions in Assumptions A.3 and A.4 would have to hold uniformly over that class. What our argument shows is narrower: for fixed P_0 , the bracketing in the proof of Theorem A.2 reduces the feasible block statistics to infeasible ones computed at the true nuisance. Whether this reduction, under the two strengthenings, combines with the uniform subsampling results of Romano and Shaikh (2010) to give uniform coverage is a conjecture that we do not pursue.

APPENDIX B: PROOFS

This appendix proves the results of Appendix A for the set estimator $\hat{\Theta}_I$ of Definition 2.1. Appendix B.1 states high-level conditions, adapted from Chernozhukov et al. (2007) and Chernozhukov et al. (2013), and three auxiliary lemmas. Appendix B.2 proves the lemmas, Appendix B.3 proves Theorem A.1, Lemma A.1 and Theorem A.2, and Appendix B.4 verifies the conditions for the examples.

B.1. Auxiliary statements

All statements below are for the K -fold cross-fitted criterion

$$\begin{aligned} Q_N(\theta, \hat{\xi}) &:= \left\| \frac{1}{N} \sum_{k=1}^K \sum_{i \in J_k} g(Z_i; \theta, \hat{\xi}^{(k)}(Z_i)) \right\|_+^2, \\ Q_N(\theta, \xi) &:= \left\| \frac{1}{N} \sum_{i=1}^N g(Z_i; \theta, \xi(Z_i)) \right\|_+^2 \end{aligned} \quad (\text{B.1})$$

for a deterministic $\xi \in \Xi$, where $\hat{\xi}^{(k)}$ is built from $J_k^c := \{1, \dots, N\} \setminus J_k$ only. The number of folds $K \geq 2$ is fixed and, for simplicity, divides N and the block size b , so that the weights $1/K$ below equal the fold shares. Every empirical-process bound below is first established fold by fold, conditionally on J_k^c . Given J_k^c , $\hat{\xi}^{(k)}$ is a fixed function, which lies in Ξ_N on the event $\mathcal{B}_N$, and the n observations of J_k are i.i.d. and independent of it. A bound that holds with conditional probability at least $1 - p$ on the J_k^c -measurable event $\mathcal{B}_N^{(k)} := \{\hat{\xi}^{(k)} \in \Xi_N\}$ holds unconditionally with probability at least $(1 - p)(1 - \phi_N)$, since $P(A) \geq E[\mathbf{1}\{\mathcal{B}_N^{(k)}\}P(A | J_k^c)]$, and the K fold-level bounds are combined by the triangle inequality at the cost of a factor K absorbed into the constants. To keep the notation light we suppress the fold index and write $\hat{\xi}_i$ for $\hat{\xi}^{(k(i))}(Z_i)$. We write $A_N \lesssim_P B_N$ when $A_N = O_P(B_N)$.

Expectations of functions of the cross-fitted nuisance are fold-conditional throughout: for a function h ,

$$E h(Z_i, \theta, \hat{\xi}_i) := \frac{1}{K} \sum_{k=1}^K E[h(Z, \theta, \hat{\xi}^{(k)}) | J_k^c], \quad (\text{B.2})$$

$$\mathbb{G}_N h(Z_i, \theta, \hat{\xi}_i) := \sqrt{N} (E_N - E) h(Z_i, \theta, \hat{\xi}_i), \quad (\text{B.3})$$

so that $N Q_N(\theta, \hat{\xi}) = \|\mathbb{G}_N g(Z_i; \theta, \hat{\xi}_i) + \sqrt{N} \mathbb{E} g(Z_i; \theta, \hat{\xi}_i)\|_+^2$ holds by definition, and on $\mathcal{B}_N$ the fold-conditional bias obeys $\|\mathbb{E} g(Z_i; \theta, \hat{\xi}_i) - \mathbb{E} g(Z_i; \theta, \xi_0)\| \leq \bar{s}_N$ by Assumption A.3(e), since each of the K conditional biases does. The high-level conditions below are stated for the cross-fitted criterion $Q_N(\theta, \hat{\xi})$ itself, not for a deterministic nuisance: with $K \geq 2$ folds the sample is not i.i.d. conditionally on $\hat{\xi}$, so a statement for deterministic $\xi \in \Xi_N$ does not transfer to $\hat{\xi}$ by conditioning, and the argument is carried out fold by fold.

Assumptions B.1 to B.3 below are high-level conditions on the population and sample criteria. Lemma B.1 derives the rate from them, Lemma B.3 verifies Assumptions B.2 and B.3 from the conditions of Appendix A, and the proof of Theorem A.1 verifies Assumption B.1.

Rates are stated in the Hausdorff distance (2.6). As in Section 2.1, all random quantities below are assumed measurable, including suprema and infima over subsets of Θ .

ASSUMPTION B.1. (CONSISTENCY OF THE CRITERION) *We impose the following:*

- (a) (Regularity of $Q(\theta, \xi_0)$.) $Q(\theta, \xi_0)$ is non-negative, and for every $\epsilon > 0$ there exists $\varpi(\epsilon) > 0$ such that

$$\inf_{\theta \in \Theta \setminus \Theta_I^\epsilon} Q(\theta, \xi_0) \geq \varpi(\epsilon) > 0. \quad (\text{B.4})$$

- (b) (Fast convergence on Θ_I .) Uniformly on Θ_I ,

$$\sup_{\theta \in \Theta_I} N Q_N(\theta, \xi_0) = O_P(1). \quad (\text{B.5})$$

- (c) (Slow convergence on Θ .) Uniformly on Θ ,

$$\sup_{\theta \in \Theta} (Q(\theta, \xi_0) - Q_N(\theta, \xi_0))_+ = O_P(N^{-1/2}). \quad (\text{B.6})$$

- (d) (Separation away from Θ_I .) There exist $\omega, \kappa > 0$ such that for any $p \in (0, 1)$ there exist D_p, N_p with, for all $N \geq N_p$,

$$\mathbb{P}\left(Q_N(\theta, \hat{\xi}) \geq \kappa [d(\theta, \Theta_I) \wedge \omega]^2 \quad \forall \theta : d(\theta, \Theta_I) \geq (D_p/N)^{1/2}\right) \geq 1 - p. \quad (\text{B.7})$$

ASSUMPTION B.2. (NO EFFECT OF FIRST-STAGE ESTIMATION ERROR) *We impose the following:*

- (a) (Slow convergence on Θ .) For any $p > 0$ there exist r_p, N_p such that for all $N \geq N_p$,

$$\mathbb{P}\left(\sqrt{N} \sup_{\theta \in \Theta} |Q_N(\theta, \hat{\xi}) - Q_N(\theta, \xi_0)| \leq r_p\right) \geq 1 - p.$$

- (b) (Fast convergence on Θ_I .) For any $p, \epsilon > 0$ there exists $N_{p, \epsilon}$ such that for all $N \geq N_{p, \epsilon}$,

$$\mathbb{P}\left(N \sup_{\theta \in \Theta_I} |Q_N(\theta, \hat{\xi}) - Q_N(\theta, \xi_0)| \leq \epsilon\right) \geq 1 - p.$$

ASSUMPTION B.3. (FAST CONVERGENCE AT BLOCK SIZE b) Let $\epsilon_N := \sqrt{(1 \vee c_N)/N}$ be the Hausdorff rate delivered by Theorem A.1 at the preliminary level c_N of Definition 2.2, up to a constant. For the recommended level $c_N = \log N$ this is $\epsilon_N = \sqrt{\log N/N}$. Let $b = b_N \rightarrow \infty$ with $b c_N/N \rightarrow 0$, and let $Q_{j,b}(\theta, \hat{\xi})$ be the criterion computed on a

block j of size b , balanced across folds, with the cross-fitted nuisance of the full sample rather than one re-estimated on the block. For every fixed $C_0 > 0$ and any $p, \epsilon > 0$ there exists N_{p, ϵ, C_0} such that for all $N \geq N_{p, \epsilon, C_0}$ and every block j ,

$$\mathbb{P}\left(b \sup_{\theta \in \Theta_I^{C_0 \epsilon N}} |Q_{j,b}(\theta, \hat{\xi}) - Q_{j,b}(\theta, \xi_0)| \leq \epsilon\right) \geq 1 - p.$$

The next three lemmas provide the key technical tools. The first establishes coverage, consistency, and the rate of convergence under the high-level assumptions.

LEMMA B.1. (COVERAGE, CONSISTENCY, AND RATE OF CONVERGENCE) *Let $\hat{c}/N \rightarrow_p 0$ and*

$$\mathbb{P}\left(\sup_{\theta \in \Theta_I} N Q_N(\theta, \hat{\xi}) \leq \hat{c}\right) = 1 - o(1).$$

Then:

- (i) *Assumption B.1(a),(c) and Assumption B.2(a) imply $\Theta_I \subseteq \hat{\Theta}_I$ with probability approaching one and $d_H(\hat{\Theta}_I, \Theta_I) = o_P(1)$.*
- (ii) *Assumption B.1(d) implies $d_H(\hat{\Theta}_I, \Theta_I) = O_P(\sqrt{(1 \vee \hat{c})/N})$.*
- (iii) *If Assumption B.1(b) and Assumption B.2(b) hold, one can choose $\hat{c} = O_P(1)$.*

The second lemma controls the concentration of the empirical process of the estimated moments around their population counterparts.

LEMMA B.2. (CONCENTRATION OF ESTIMATED MOMENTS) *Suppose Assumptions A.3 and A.4 hold. Let a block be a deterministic set of $m = m_N \rightarrow \infty$ indices with m/K from every fold, and let $\mathbb{G}_m$ be the empirical process of the block with the full-sample cross-fitted nuisance $\hat{\xi}_i$; $m = N$ gives the full sample. For every $p \in (0, 1)$ there exist $C_p < \infty$ and N_p , depending on (K, a, v, c, c_1, p) only and not on the block, such that for all $N \geq N_p$, with $\zeta_m := r'_N \sqrt{\log m} + m^{-1/2+1/c} \log m$,*

$$\mathbb{P}\left(\sup_{\theta \in \Theta} \left\| \mathbb{G}_m \left\{ g(Z_i; \theta, \hat{\xi}_i) - g(Z_i; \theta, \xi_0) \right\} \right\| \leq C_p \zeta_m \right) \geq 1 - p. \quad (\text{B.8})$$

The third lemma shows that the high-level Assumptions B.2 and B.3 follow from the conditions stated in Appendix A.

LEMMA B.3. (SUFFICIENT CONDITIONS FOR GENERAL MOMENT PROBLEMS) *Assumptions A.3 and A.4 and (A.2) imply Assumption B.2, and together with Assumption A.5 and a block size $b \rightarrow \infty$ with $b c_N / N \rightarrow 0$ they imply Assumption B.3.*

B.2. Proofs of auxiliary lemmas

Proof of Lemma B.1: The proof uses two elementary inequalities: for real functions f and h on a set A ,

$$\sup_A f \leq \sup_A h + \sup_A (f - h)_+, \quad \inf_A f \geq \inf_A h - \sup_A (h - f)_+. \quad (\text{B.9})$$

Step 1: containment. The event $\{\sup_{\theta \in \Theta_I} N Q_N(\theta, \hat{\xi}) \leq \hat{c}\}$ is the event $\{\Theta_I \subseteq \hat{\Theta}_I\}$, so the

level condition of the lemma is $P(\Theta_I \subseteq \widehat{\Theta}_I) \rightarrow 1$. *Step 2: convergence without rate.* Fix

$\epsilon > 0$. We show $P(d_H(\widehat{\Theta}_I, \Theta_I) \leq \epsilon) \rightarrow 1$. By Assumption B.1(a), there exists $\varpi(\epsilon) > 0$ such that

$$\inf_{\theta \in \Theta \setminus \Theta_I^\epsilon} Q(\theta, \xi_0) \geq \varpi(\epsilon) > 0.$$

We chain inequalities using (B.9):

$$\begin{aligned} \sup_{\theta \in \widehat{\Theta}_I} Q(\theta, \xi_0) &\leq \sup_{\theta \in \widehat{\Theta}_I} Q_N(\theta, \xi_0) + \sup_{\theta \in \widehat{\Theta}_I} (Q(\theta, \xi_0) - Q_N(\theta, \xi_0))_+ && \text{(by (B.9))} \\ &\leq \sup_{\theta \in \widehat{\Theta}_I} Q_N(\theta, \xi_0) + O_P(1/\sqrt{N}) && \text{(by (B.6))} \\ &\leq \sup_{\theta \in \widehat{\Theta}_I} Q_N(\theta, \widehat{\xi}) + \sup_{\theta \in \widehat{\Theta}_I} |Q_N(\theta, \xi_0) - Q_N(\theta, \widehat{\xi})| + O_P(1/\sqrt{N}) && \text{(by (B.9))} \\ &\leq \widehat{c}/N + O_P(N^{-1/2}) + O_P(N^{-1/2}) = o_P(1). && \text{(Assumption B.2(a) and } \widehat{c}/N \rightarrow_p 0) \end{aligned}$$

Hence, by (B.4), $P(\widehat{\Theta}_I \subseteq \Theta_I^\epsilon) = 1 - o(1)$. *Step 3: convergence rate.* Fix $p \in (0, 1)$,

let κ, ω be as in Assumption B.1(d), let $D_{p/3}$ be as in (B.7), and set $\epsilon_{N,p} := (2 \max\{D_{p/3}, \widehat{c}/\kappa\}/N)^{1/2}$. We show that $P(d_H(\widehat{\Theta}_I, \Theta_I) \leq \epsilon_{N,p}) \geq 1 - p$ for all large N .

Since $\widehat{c}/N \rightarrow_p 0$, there exists $N_{p/3}^A$ such that for all $N \geq N_{p/3}^A$,

$$P(\mathcal{A}_N) := P(\widehat{c}/(N\kappa) < \omega^2/8) \geq 1 - p/3.$$

On $\mathcal{A}_N$ and for N large,

$$\epsilon_{N,p}^2 = \frac{2 \max\{D_{p/3}, \widehat{c}/\kappa\}}{N} \leq \max\left\{\frac{2D_{p/3}}{N}, \frac{\omega^2}{4}\right\} \leq \frac{\omega^2}{4}, \quad \epsilon_{N,p} \geq \sqrt{D_{p/3}/N},$$

so Assumption B.1(d) applies at radius $\epsilon_{N,p}$.

Define

$$\mathcal{D}_{\widehat{\xi}} := \left\{ \inf_{\theta: d(\theta, \Theta_I) \geq \epsilon_{N,p}} N Q_N(\theta, \widehat{\xi}) \geq N\kappa(\epsilon_{N,p} \wedge \omega)^2 \right\}.$$

Since $\epsilon_{N,p} \geq (D_{p/3}/N)^{1/2}$ on $\mathcal{A}_N$, Assumption B.1(d) gives $P(\mathcal{D}_{\widehat{\xi}} \cap \mathcal{A}_N) \geq 1 - 2p/3$ for $N \geq N_{p/3}$. Therefore, with probability at least $1 - 2p/3$ and for all $N \geq N_{p/3} \vee N_{p/3}^A$,

$$\inf_{\theta: d(\theta, \Theta_I) \geq \epsilon_{N,p}} N Q_N(\theta, \widehat{\xi}) \geq N\kappa(\epsilon_{N,p} \wedge \omega)^2 = N\kappa\epsilon_{N,p}^2 = 2 \max\{\kappa D_{p/3}, \widehat{c}\} > \widehat{c}, \quad (\text{B.10})$$

since $D_{p/3} > 0$, which is without loss of generality in Assumption B.1(d). Every θ with $d(\theta, \Theta_I) \geq \epsilon_{N,p}$ is therefore excluded from $\widehat{\Theta}_I = \{N Q_N(\theta, \widehat{\xi}) \leq \widehat{c}\}$, so $P(\widehat{\Theta}_I \subseteq \Theta_I^{\epsilon_{N,p}}) \geq 1 - 2p/3$ for large N . Together with Step 1, this gives $d_H(\widehat{\Theta}_I, \Theta_I) \leq \epsilon_{N,p}$ with probability at least $1 - p$ for large N , and $\epsilon_{N,p} \lesssim \sqrt{(1 \vee \widehat{c})/N}$. *Step 4: choice $\widehat{c} = O_P(1)$.* That some

$\widehat{c} = O_P(1)$ satisfies the level condition follows from

$$\begin{aligned} \sup_{\theta \in \Theta_I} N Q_N(\theta, \widehat{\xi}) &\leq \sup_{\theta \in \Theta_I} N Q_N(\theta, \xi_0) + \sup_{\theta \in \Theta_I} N |Q_N(\theta, \widehat{\xi}) - Q_N(\theta, \xi_0)| \\ &\leq O_P(1) + \sup_{\theta \in \Theta_I} N |Q_N(\theta, \widehat{\xi}) - Q_N(\theta, \xi_0)| \quad (\text{Assumption B.1(b)}) \\ &\leq O_P(1) + o_P(1), \quad (\text{Assumption B.2(b)}) \end{aligned}$$

so any $\widehat{c}$ that exceeds the $O_P(1)$ infeasible statistic with probability approaching one satisfies it, for instance $\widehat{c} := \sup_{\theta \in \Theta_I} N Q_N(\theta, \widehat{\xi})$ itself, which is infeasible, or a slowly diverging deterministic sequence, which is feasible but not $O_P(1)$. The subsampling level $\widehat{c}_\tau$ of Definition 2.2 covers only with probability $1 - \tau + o(1)$ and does not satisfy the level condition. ■

Proof of Lemma B.2: Fix a fold k and condition on the observations outside J_k , so that $\widehat{\xi}^{(k)}$ is a fixed function, which lies in Ξ_N on $\mathcal{B}_N$. The observations of the block that belong to J_k are m/K i.i.d. draws independent of $\widehat{\xi}^{(k)}$, and for the full sample $m = N$. We bound the empirical process of these m/K observations and combine the K bounds at the end. Write $\widehat{\xi}$ for $\widehat{\xi}^{(k)}$. Consider the function class

$$\mathcal{F}_2 = \left\{ g_l(\cdot; \theta, \widehat{\xi}) - g_l(\cdot; \theta, \xi_0) : l = 1, \dots, L, \theta \in \Theta \right\} \subset \mathcal{F}_{\widehat{\xi}} - \mathcal{F}_{\xi_0}.$$

Let $F_2 := F_{\widehat{\xi}} + F_{\xi_0}$ be an envelope for $\mathcal{F}_2$. This envelope satisfies the requirements of Lemma 6.2 of Chernozhukov et al. (2018), since $\|F_2\|_{P,c} \leq \|F_{\widehat{\xi}}\|_{P,c} + \|F_{\xi_0}\|_{P,c} \leq 2c_1$ and

$$\log \sup_{\tilde{Q}} \mathcal{N} \left(\epsilon \|F_2\|_{\tilde{Q},2}, \mathcal{F}_2, \|\cdot\|_{\tilde{Q},2} \right) \leq 2v \log(2aL/\epsilon),$$

where the factor L accounts for the L coordinates and is absorbed into the constants. On the event $\mathcal{B}_N$, the variance of the difference class is controlled as

$$\begin{aligned} \sup_{\theta \in \Theta} \mathbb{E} \left[\{g_l(Z; \theta, \widehat{\xi}) - g_l(Z; \theta, \xi_0)\}^2 \mid J_k^c \right] &\leq \sup_{\theta \in \Theta, \xi \in \Xi_N} \mathbb{E} \|g(Z; \theta, \xi) - g(Z; \theta, \xi_0)\|^2 \\ &\leq (r'_N)^2. \end{aligned}$$

Let $m' = m/K \rightarrow \infty$ be the number of block observations in the fold and take the variance bound $\sigma_m := r'_N \vee (m')^{-1/2+1/c}$. Lemma 6.2 of Chernozhukov et al. (2018) requires $\sup_{f \in \mathcal{F}_2} \mathbb{E} f^2 \leq \sigma_m^2 \leq \|F_2\|_{P,2}^2$. The first inequality is the variance bound above. For the second, we may take $F_{\xi} \geq 1$, since enlarging an envelope does not affect Assumption A.4(a) or (b); then $\|F_2\|_{P,2} \geq 1$, while $\sigma_m \rightarrow 0$. We apply Lemma 6.2 conditionally on J_k^c , with the class $\mathcal{F}_2$, envelope F_2 , variance bound σ_m^2 and the constants $a \geq e$, $v \geq 1$ of Assumption A.4(b), and use its tail bound with $t = p^{-2/c}$. With conditional probability at least $1 - p$,

$$\begin{aligned} \sup_{f \in \mathcal{F}_2} |\mathbb{G}_{m'} f| &\leq K_p \left(\sigma_m \sqrt{v \log(a \|F_2\|_{P,2} / \sigma_m)} \right. \\ &\quad \left. + v (m')^{-1/2+1/c} \|F_2\|_{P,c} \log(a \|F_2\|_{P,2} / \sigma_m) \right). \end{aligned}$$

Since $\sigma_m \geq (m')^{-1/2+1/c}$, the logarithms are at most a constant times $\log m$, so the right-hand side is $\lesssim \zeta_m$ with constants depending only on (K, a, v, c, c_1, p) and not on the block or the fold. The additional terms of the tail bound, $\sigma_m \sqrt{t}$, $(m')^{-1/2} \|\max_i F_2(Z_i)\|_{P,c} \sqrt{t}$

and $(m')^{-1/2} \|\max_i F_2(Z_i)\|_{P,2} t$, are of the same order, since $\|\max_{i \leq m'} F_2(Z_i)\|_{P,c} \leq (m')^{1/c} \|F_2\|_{P,c}$. The bound is integrated over the conditioning at the cost of the factor $1 - \phi_N$ as explained at the start of this appendix, and the K fold bounds are combined by the triangle inequality, since $\mathbb{G}_m$ is the average of the K fold processes scaled by $\sqrt{K}$. This gives

$$\sup_{\theta \in \Theta} \left\| \mathbb{G}_m \left\{ g(Z_i; \theta, \hat{\xi}_i) - g(Z_i; \theta, \xi_0) \right\} \right\| \lesssim_P \zeta_m, \quad (\text{B.11})$$

which is $o_P(1)$ for $m = N$ by Assumption A.4(c), and for every $m \leq N$ with $m \rightarrow \infty$ since $r'_N \sqrt{\log m} \leq r'_N \sqrt{\log N}$. $\blacksquare$

Proof of Lemma B.3: The proof bounds the difference between the feasible and the infeasible criterion, first on the full sample, which gives Assumption B.2, and then on a block of size b , which gives Assumption B.3. Fix $p \in (0, 1)$, let $\mathcal{B}_N$ be the event of Assumption A.3(b), and let $\mathcal{E}_{1,N}$ and $\mathcal{E}_{2,N}$ be the events, provided with a common constant $C_{p/4}$ by Lemma B.2 and (A.2), on which

$$\mathbb{P}(\mathcal{E}_{1,N}) := \mathbb{P} \left(\sup_{\theta \in \Theta} \left\| \mathbb{G}_N \left\{ g(Z_i; \theta, \hat{\xi}_i) - g(Z_i; \theta, \xi_0) \right\} \right\| \leq C_{p/4} \zeta_N \right) \geq 1 - p/4,$$

$$\mathbb{P}(\mathcal{E}_{2,N}) := \mathbb{P} \left(\sup_{\theta \in \Theta} \left\| \mathbb{G}_N g(Z_i; \theta, \xi_0) \right\| \leq C_{p/4} \right) \geq 1 - p/4.$$

Here ζ_N is ζ_m of Lemma B.2 at $m = N$, so $\mathcal{E}_{1,N}$ has the stated probability by that lemma, and $\zeta_N = o(1)$ by Assumption A.4(c). On $\mathcal{B}_N \cap \mathcal{E}_{1,N} \cap \mathcal{E}_{2,N}$, we decompose the criterion function difference. Setting $x := \mathbb{E}_N g(Z_i; \theta, \hat{\xi}_i)$ and $y := \mathbb{E}_N g(Z_i; \theta, \xi_0)$,

$$|Q_N(\theta, \hat{\xi}) - Q_N(\theta, \xi_0)| = \left| \|x\|_+^2 - \|y\|_+^2 \right| \leq \|x - y\|^2 + 2\|x - y\| \|y\|_+,$$

where the inequality uses $|||a||_+^2 - |||b||_+^2| \leq \|a - b\|^2 + 2\|a - b\| \|b\|_+$, a consequence of the triangle inequality for $\|\cdot\|_+$. The two factors are bounded as follows:

$$\begin{aligned} \|x - y\| &= \left\| \mathbb{E}[g(Z_i; \theta, \hat{\xi}_i) - g(Z_i; \theta, \xi_0)] + \mathbb{G}_N[g(Z_i; \theta, \hat{\xi}_i) - g(Z_i; \theta, \xi_0)]/\sqrt{N} \right\| \\ &\leq \bar{s}_N + C_{p/4} N^{-1/2} \zeta_N, \\ \|y\|_+ &= \left\| \mathbb{G}_N g(Z_i; \theta, \xi_0)/\sqrt{N} + \mathbb{E} g(Z_i; \theta, \xi_0) \right\|_+. \end{aligned}$$

Verification of Assumption B.2(a). Taking

$$r_p = (2C_{p/4})^2 + 4C_{p/4} \left(C_{p/4} + \sup_{\theta \in \Theta} \|\mathbb{E} g(Z_i; \theta, \xi_0)\| \right),$$

and using $\sqrt{N} \bar{s}_N + C_{p/4} \zeta_N \leq 2C_{p/4}$ for N large, we obtain, on $\mathcal{B}_N \cap \mathcal{E}_{1,N} \cap \mathcal{E}_{2,N}$ and hence with probability at least $1 - p$,

$$\sqrt{N} \sup_{\theta \in \Theta} |Q_N(\theta, \hat{\xi}) - Q_N(\theta, \xi_0)| \leq r_p.$$

Verification of Assumption B.2(b). On Θ_I , $\|\mathbb{E} g(Z_i; \theta, \xi_0)\|_+ = 0$ by definition of the identified set, so

$$\sup_{\theta \in \Theta_I} \sqrt{N} \|y\|_+ \leq \sup_{\theta \in \Theta_I} \left\| \mathbb{G}_N g(Z_i; \theta, \xi_0) \right\| \leq C_{p/4},$$

and hence, on $\mathcal{B}_N \cap \mathcal{E}_{1,N} \cap \mathcal{E}_{2,N}$,

$$N \sup_{\theta \in \Theta_I} |Q_N(\theta, \hat{\xi}) - Q_N(\theta, \xi_0)| \leq (\sqrt{N} \bar{s}_N + C_{p/4} \zeta_N)^2 + 2(\sqrt{N} \bar{s}_N + C_{p/4} \zeta_N) C_{p/4} \leq \epsilon$$

for N large, since $\sqrt{N} \bar{s}_N \rightarrow 0$ and $\zeta_N \rightarrow 0$. This gives Assumption B.2(b).

Verification of Assumption B.3. Fix a block j of size b , balanced across folds, and let x and y now denote the block averages $E_{j,b} g(Z_i; \theta, \hat{\xi}_i)$ and $E_{j,b} g(Z_i; \theta, \xi_0)$. Every block draws b/K observations from each fold, and those in fold k are i.i.d. and independent of $\hat{\xi}^{(k)}$, so Lemma B.2 applies at $m = b$ with the full-sample nuisance: with probability at least $1 - p/4$,

$$\sup_{\theta \in \Theta} \sqrt{b} \|x - y\| \leq \sqrt{b} \bar{s}_N + C_{p/4} \zeta_b,$$

with ζ_b the ζ_m of Lemma B.2 at $m = b$ and a constant that depends on the block only through b . Both terms are $o(1)$: $\sqrt{b} \bar{s}_N \leq \sqrt{N} \bar{s}_N$ by Assumption A.3(e), $r'_N \sqrt{\log b} \leq r'_N \sqrt{\log N}$ by Assumption A.4(c), and $b^{-1/2+1/c} \log b \rightarrow 0$ as $b \rightarrow \infty$. For $\theta \in \Theta_I^{C_0 \epsilon_N}$,

$$b |Q_{j,b}(\theta, \hat{\xi}) - Q_{j,b}(\theta, \xi_0)| \leq b \|x - y\|^2 + 2b \|x - y\| \|y\|_+, \quad (\text{B.12})$$

so the first term is $o(1)$. For the second factor, (A.2) applied to the b i.i.d. observations of the block gives an event of probability at least $1 - p/4$ on which $\sup_{\theta \in \Theta} \|\mathbb{G}_b g(Z_i; \theta, \xi_0)\| \leq C_{p/4}$, and there Assumption A.5 gives

$$\begin{aligned} \sup_{\theta \in \Theta_I^{C_0 \epsilon_N}} \sqrt{b} \|y\|_+ &\leq C_{p/4} + \sup_{\theta \in \Theta_I^{C_0 \epsilon_N}} \sqrt{b} \|E g(Z_i; \theta, \xi_0)\|_+ \\ &\leq C_{p/4} + C_{\max} C_0 \sqrt{b} \epsilon_N = C_{p/4} + o(1), \end{aligned}$$

because $\sqrt{b} \epsilon_N = \sqrt{b(1 \vee c_N)/N} \rightarrow 0$ by hypothesis. The second term of (B.12) is therefore the product of an $o(1)$ and an $O(1)$ factor. Thus, for any $\epsilon, p > 0$ and N large,

$$\mathbb{P} \left(\sup_{\theta \in \Theta_I^{C_0 \epsilon_N}} b |Q_{j,b}(\theta, \hat{\xi}) - Q_{j,b}(\theta, \xi_0)| \leq \epsilon \right) \geq 1 - p,$$

uniformly in j . On a block, the bias enters at scale $\sqrt{b} \bar{s}_N$ and the expansion at scale $\sqrt{b} \epsilon_N$, both smaller than their full-sample counterparts, so no condition beyond Assumption A.3(e), Assumption A.4(c) and $b c_N/N \rightarrow 0$ is needed. The same decomposition at $b = N$ shows that the full-sample analogue would require $\sqrt{N} \epsilon_N (\sqrt{N} \bar{s}_N + \zeta_N) = o(1)$, a strengthening of the bias and continuity conditions by the factor $\sqrt{1 \vee c_N}$; the subsampling construction does not need it. $\blacksquare$

B.3. Proofs of the main results

Proof of Theorem A.1: We verify that Assumption B.1(a)–(d) hold under the conditions of Appendix A, then apply Lemma B.1.

Verification of Assumption B.1(a). By Assumption A.1, for every $\epsilon > 0$,

$$\inf_{\theta \in \Theta \setminus \Theta_I^\epsilon} Q(\theta, \xi_0) = \inf_{\theta \in \Theta \setminus \Theta_I^\epsilon} \|E g(Z_i; \theta, \xi_0)\|_+^2 \geq C_{\min}^2 (\epsilon \wedge \delta_{\min})^2 > 0.$$

Verification of Assumption B.1(b). For $\theta \in \Theta_I$ we have $E g_l(Z_i; \theta, \xi_0) \leq 0$ for every l ,

and the map $v \mapsto \|v\|_+$ is coordinatewise non-decreasing. Hence, writing $\sqrt{N} \mathbb{E}_N g = \mathbb{G}_N g + \sqrt{N} \mathbb{E} g$ and dropping the non-positive second term,

$$\begin{aligned} \sup_{\theta \in \Theta_I} N Q_N(\theta, \xi_0) &= \sup_{\theta \in \Theta_I} \left\| \mathbb{G}_N g(Z_i; \theta, \xi_0) + \sqrt{N} \mathbb{E} g(Z_i; \theta, \xi_0) \right\|_+^2 \\ &\leq \sup_{\theta \in \Theta_I} \left\| \mathbb{G}_N g(Z_i; \theta, \xi_0) \right\|_+^2 = O_P(1), \end{aligned}$$

the last step by the P -Donsker property (A.2) and the continuous mapping theorem.

Verification of Assumption B.1(c). Again by the P -Donsker property,

$$\begin{aligned} \sup_{\theta \in \Theta} \sqrt{N} \left| \left\| \mathbb{E} g(Z; \theta, \xi_0) \right\|_+^2 - \left\| \mathbb{E}_N g(Z; \theta, \xi_0) \right\|_+^2 \right| \\ \leq \left(\sup_{\theta} \left\| \mathbb{E} g(Z; \theta, \xi_0) \right\|_+ + \sup_{\theta} \left\| \mathbb{E}_N g(Z; \theta, \xi_0) \right\|_+ \right) \sup_{\theta} \left\| \mathbb{G}_N g(Z; \theta, \xi_0) \right\| = O_P(1). \end{aligned}$$

Verification of Assumption B.1(d). Fix $p > 0$. By Assumption A.3(b), $P(\mathcal{B}_N) \geq 1 - p/3$ for N large, and by Lemma B.2 and (A.2) there exists $C_{p/3} > 0$ such that, for N large,

$$\begin{aligned} P(\mathcal{E}'_{1,N}) &:= P \left(\sup_{\theta \in \Theta} \left\| \mathbb{G}_N \left\{ g(Z_i; \theta, \hat{\xi}_i) - g(Z_i; \theta, \xi_0) \right\} \right\| \leq C_{p/3} \right) \geq 1 - p/3, \\ P(\mathcal{E}'_{2,N}) &:= P \left(\sup_{\theta \in \Theta} \left\| \mathbb{G}_N g(Z_i; \theta, \xi_0) \right\| \leq C_{p/3} \right) \geq 1 - p/3. \end{aligned}$$

On $\mathcal{B}_N \cap \mathcal{E}'_{1,N} \cap \mathcal{E}'_{2,N}$ and for large N , by the fold-conditional convention (B.2),

$$N Q_N(\theta, \hat{\xi}) = \left\| \mathbb{G}_N g(Z_i; \theta, \hat{\xi}_i) + \sqrt{N} \mathbb{E} g(Z_i; \theta, \hat{\xi}_i) \right\|_+^2.$$

Fix θ with $d(\theta, \Theta_I) \geq 6C_{p/3}/(C_{\min} \sqrt{N})$ and set

$$\begin{aligned} \mu_N &:= \sqrt{N} \mathbb{E} g(Z_i; \theta, \xi_0), \\ R_N &:= \mathbb{G}_N g(Z_i; \theta, \xi_0) + \mathbb{G}_N \left\{ g(Z_i; \theta, \hat{\xi}_i) - g(Z_i; \theta, \xi_0) \right\} \\ &\quad + \sqrt{N} \mathbb{E} \left[g(Z_i; \theta, \hat{\xi}_i) - g(Z_i; \theta, \xi_0) \right]. \end{aligned}$$

By Assumption A.1,

$$\|\mu_N\|_+ \geq \sqrt{N} C_{\min} (d(\theta, \Theta_I) \wedge \delta_{\min}) \geq 6C_{p/3}.$$

By $\mathcal{E}'_{1,N}, \mathcal{E}'_{2,N}$ and the bound $\sqrt{N} \bar{s}_N < C_{p/3}$ for N large, which Assumption A.3(e) gives for the fold-conditional bias on $\mathcal{B}_N$,

$$\|R_N\| \leq C_{p/3} + C_{p/3} + C_{p/3} = 3C_{p/3}.$$

Hence $\|\mu_N\|_+ \geq 2\|R_N\|$, and the triangle inequality gives

$$\|\mu_N + R_N\|_+ \geq \|\mu_N\|_+ - \|R_N\| \geq \frac{1}{2} \|\mu_N\|_+,$$

so that

$$N Q_N(\theta, \hat{\xi}) = \|\mu_N + R_N\|_+^2 \geq \frac{1}{4} \|\mu_N\|_+^2 \geq \frac{1}{4} N C_{\min}^2 [d(\theta, \Theta_I) \wedge \delta_{\min}]^2.$$

This verifies Assumption B.1(d) with $\kappa = \frac{1}{4} C_{\min}^2$, $\omega = \delta_{\min}$, and $D_p = (6C_{p/3}/C_{\min})^2$. Assumption B.2(a) holds by Lemma B.3, so Lemma B.1(i)–(ii) give containment with probability approaching one and the rate. $\blacksquare$

Proof of Lemma A.1: Step 1 shows that the cross-fitted criterion vanishes on a contraction $\Theta_I^{-\widehat{\epsilon}_N}$ with radius $\widehat{\epsilon}_N = O_P(N^{-1/2})$, which is the degeneracy property of Definition A.1. Step 2 derives the rate from it and the choice of contour level $\widehat{\mathcal{C}}'$.

Step 1: degeneracy on an inner contraction. Fix $\epsilon \in (0, \delta]$ and $\theta \in \Theta_I^{-\epsilon}$. By the fold-conditional convention (B.2),

$$N Q_N(\theta, \widehat{\xi}) = \left\| \mathbb{G}_N g(Z_i; \theta, \widehat{\xi}_i) + \sqrt{N} \mathbb{E} g(Z_i; \theta, \widehat{\xi}_i) \right\|_+^2,$$

and componentwise, for $l = 1, \dots, L$,

$$\begin{aligned} \mathbb{G}_N g_l(\theta, \widehat{\xi}) + \sqrt{N} \mathbb{E} g_l(\theta, \widehat{\xi}) &= \underbrace{\mathbb{G}_N g_l(\theta, \xi_0) + \sqrt{N} \mathbb{E} g_l(\theta, \xi_0)}_{(A)} \\ &\quad + \underbrace{\mathbb{G}_N [g_l(\theta, \widehat{\xi}) - g_l(\theta, \xi_0)]}_{(B)} + \underbrace{\sqrt{N} \mathbb{E} [g_l(\theta, \widehat{\xi}) - g_l(\theta, \xi_0)]}_{(C)}, \end{aligned}$$

where (A) is the term at the true nuisance, (B) the empirical process of the nuisance estimation error, and (C) the fold-conditional bias.

By (A.8), $\mathbb{E} g_l(\theta, \xi_0) \leq -C(\epsilon \wedge \delta) = -C\epsilon$ for $\theta \in \Theta_I^{-\epsilon}$, and by (A.2) the oracle process is bounded,

$$\mathbb{M}_N := \sup_{\theta \in \Theta, 1 \leq l \leq L} |\mathbb{G}_N g_l(Z_i; \theta, \xi_0)| = O_P(1),$$

so (A) $\leq \mathbb{M}_N - \sqrt{N} C\epsilon$. For (B), Lemma B.2 gives $\sup_{\theta} \max_l |(B)| \lesssim_P \zeta_N = o_P(1)$ by Assumption A.4(c). For (C), Assumption A.3(e) gives $\sup_{\theta} \max_l |(C)| \leq \sqrt{N} \bar{s}_N = o(1)$ on $\mathcal{B}_N$. Hence the event

$$\mathcal{H}_N := \left\{ \sup_{\theta \in \Theta} \max_l (|(B)| + |(C)|) \leq 1 \right\}$$

has probability approaching one. Choose the radius

$$\widehat{\epsilon}_N := \frac{2}{C\sqrt{N}} (\mathbb{M}_N + 1),$$

which is $O_P(N^{-1/2})$ and at most δ with probability approaching one. On $\mathcal{H}_N \cap \{\widehat{\epsilon}_N \leq \delta\}$, for every $\theta \in \Theta_I^{-\widehat{\epsilon}_N}$ and every l ,

$$\mathbb{G}_N g_l(\theta, \widehat{\xi}) + \sqrt{N} \mathbb{E} g_l(\theta, \widehat{\xi}) \leq \mathbb{M}_N - \sqrt{N} C\widehat{\epsilon}_N + 1 = -(\mathbb{M}_N + 1) \leq 0,$$

so all positive parts vanish and $Q_N(\theta, \widehat{\xi}) = 0 = \inf_{\Theta} Q_N(\cdot, \widehat{\xi})$ for all $\theta \in \Theta_I^{-\widehat{\epsilon}_N}$. Consequently, with probability approaching one, $\Theta_I^{-\widehat{\epsilon}_N} \subseteq C_N(\widehat{\mathcal{C}}'; \widehat{\xi})$ for any level $\widehat{\mathcal{C}}' \geq 0$, and by the second part of (A.8), $d_H(\Theta_I^{-\widehat{\epsilon}_N}, \Theta_I) \leq M\widehat{\epsilon}_N = O_P(N^{-1/2})$, which together is Definition A.1 with $\Theta_N = \Theta_I^{-\widehat{\epsilon}_N}$.

Step 2: rate from degeneracy and choice of $\widehat{\mathcal{C}}'$. Let $\widehat{\mathcal{C}}' = O_P(1)$ obey (A.9) and set $\widehat{\mathcal{C}} := \max\{\widehat{\mathcal{C}}', \sup_{\theta \in \Theta_I} N Q_N(\theta, \widehat{\xi})\}$, which is $O_P(1)$ by Assumption B.1(b) and Assumption B.2(b), both supplied by Lemma B.3 and the proof of Theorem A.1, and satisfies the level condition of Theorem A.1 by construction. By Step 1 there is $\widehat{\epsilon}_N = O_P(N^{-1/2})$ such that, with probability approaching one, $Q_N(\theta, \widehat{\xi}) = 0$ on $\Theta_N := \Theta_I^{-\widehat{\epsilon}_N}$, hence

$$\Theta_N \subseteq C_N(\widehat{\mathcal{C}}'; \widehat{\xi}) \subseteq C_N(\widehat{\mathcal{C}}; \widehat{\xi}) \quad \text{with probability approaching one.}$$

By the second part of (A.8), applicable since $\hat{\epsilon}_N \leq \delta$ with probability approaching one, $d_H(\Theta_N, \Theta_I) \leq M\hat{\epsilon}_N = O_P(N^{-1/2})$. By Theorem A.1 and $\hat{c} = O_P(1)$, $d_H(C_N(\hat{c}; \hat{\xi}), \Theta_I) = O_P(N^{-1/2})$. The sandwich gives both directions,

$$\begin{aligned} \sup_{\theta \in C_N(\hat{c}; \hat{\xi})} d(\theta, \Theta_I) &\leq \sup_{\theta \in C_N(\hat{c}; \hat{\xi})} d(\theta, \Theta_I), \\ \sup_{\theta \in \Theta_I} d(\theta, C_N(\hat{c}; \hat{\xi})) &\leq \sup_{\theta \in \Theta_I} d(\theta, \Theta_N), \end{aligned}$$

and both right-hand sides are $O_P(N^{-1/2})$. Hence $d_H(C_N(\hat{c}; \hat{\xi}), \Theta_I) = O_P(N^{-1/2})$. ■

Proof of Theorem A.2: Step 1 brackets each feasible block statistic between two statistics that use ξ_0 and depend only on the data of their own block, Step 2 shows that the empirical distribution functions of the brackets concentrate, and Step 3 passes to the limit law $\mathcal{T}$ and derives part (1); part (2) follows.

Step 1: two-sided bounds. Let $\epsilon_N = \sqrt{(1 \vee c_N)/N}$ be the radius of Assumption B.3, and let $C_0 := 2\sqrt{2}/C_{\min}$ be the constant with which Step 3 of the proof of Lemma B.1 delivers the rate at the level c_N : for N large, $\epsilon_{N,p}^2 = 2c_N/(\kappa N)$ there, with $\kappa = C_{\min}^2/4$, so $P(d_H(C_N(c_N; \hat{\xi}), \Theta_I) \leq C_0\epsilon_N) \rightarrow 1$. Assumptions A.7 and B.3 are stated for every fixed multiple of ϵ_N , so they apply at the radius $C_0\epsilon_N$. For a subsample j (a block of size b ; see Step 2), define

$$\bar{T}_{j,b,\xi} := \sup_{\theta \in \Theta_I^{C_0\epsilon_N}} bQ_{j,b}(\theta, \xi), \quad \underline{T}_{j,b,\xi} := \inf_{S \in \mathcal{K}_N} \sup_{\theta \in S} bQ_{j,b}(\theta, \xi),$$

where $\mathcal{K}_N := \{S \subset \Theta : S \neq \emptyset, S \text{ compact}, d_H(S, \Theta_I) \leq C_0\epsilon_N\}$. Under the continuity hypothesis, the set $\mathcal{K}_N$ is compact under the Hausdorff metric, and since $S \mapsto \sup_{\theta \in S} Q_{j,b}(\theta, \xi)$ is continuous, the infimum is attained at some $\Theta_b^* \in \mathcal{K}_N$; under (A.14), attainment is not used.

Let $\mathcal{A}_N := \{d_H(C_N(c_N; \hat{\xi}), \Theta_I) \leq C_0\epsilon_N\}$. The deterministic level $c_N \rightarrow \infty$ with $c_N/N \rightarrow 0$ satisfies the level condition of Lemma B.1 by Step 4 of its proof, so $P(\mathcal{A}_N) \rightarrow 1$ as just explained. On $\mathcal{A}_N \cap \mathcal{B}_N$ we may take $S = C_N(c_N; \hat{\xi})$ itself, which is compact under the continuity hypothesis, non-empty, and satisfies $d_H(S, \Theta_I) \leq C_0\epsilon_N$, so $S \in \mathcal{K}_N$ and $S \subseteq \Theta_I^{C_0\epsilon_N}$. Hence

$$\underline{T}_{j,b,\hat{\xi}} \leq \hat{T}_{j,b} := \sup_{\theta \in C_N(c_N; \hat{\xi})} bQ_{j,b}(\theta, \hat{\xi}) \leq \bar{T}_{j,b,\hat{\xi}}.$$

Fix $\epsilon > 0$ and let E_j denote the event $\sup_{\theta \in \Theta_I^{C_0\epsilon_N}} b|Q_{j,b}(\theta, \hat{\xi}) - Q_{j,b}(\theta, \xi_0)| \leq \epsilon$. On $\mathcal{A}_N \cap E_j$,

$$\underline{T}_{j,b,\xi_0} - \epsilon \leq \underline{T}_{j,b,\hat{\xi}} \leq \hat{T}_{j,b} \leq \bar{T}_{j,b,\hat{\xi}} \leq \bar{T}_{j,b,\xi_0} + \epsilon.$$

Because every block draws b/K observations from each fold, the observations of block j in fold k are i.i.d. and independent of $\hat{\xi}^{(k)}$, so Lemma B.3 applies to block j with the full-sample nuisance and gives $P(E_j^c) \rightarrow 0$ for each j , with a bound that depends on j only through b . Let $\hat{\pi}_N := B_N^{-1} \sum_j \mathbf{1}\{E_j^c\}$ be the fraction of blocks on which the two-sided bound may fail. Then $E\hat{\pi}_N \rightarrow 0$ and $\hat{\pi}_N = o_P(1)$ by Markov's inequality. No union bound over the $B_N \rightarrow \infty$ blocks is needed.

Define the empirical cdfs over the B_N subsamples (blocks):

$$\begin{aligned}\widehat{G}(x) &:= \frac{1}{B_N} \sum_{j=1}^{B_N} \mathbf{1}\{\widehat{T}_{j,b} \leq x\}, \\ \underline{G}(x) &:= \frac{1}{B_N} \sum_{j=1}^{B_N} \mathbf{1}\{\underline{T}_{j,b,\xi_0} \leq x\}, \\ \overline{G}(x) &:= \frac{1}{B_N} \sum_{j=1}^{B_N} \mathbf{1}\{\overline{T}_{j,b,\xi_0} \leq x\}.\end{aligned}$$

From the two-sided bound, since $\widehat{T}_{j,b} \leq x$ whenever $\overline{T}_{j,b,\xi_0} \leq x - \epsilon$ and $\underline{T}_{j,b,\xi_0} \leq x + \epsilon$ whenever $\widehat{T}_{j,b} \leq x$ on E_j , on $\mathcal{A}_N$ and for any fixed $\epsilon > 0$ and every x ,

$$\overline{G}(x - \epsilon) - \widehat{\pi}_N \leq \widehat{G}(x) \leq \underline{G}(x + \epsilon) + \widehat{\pi}_N.$$

The compactness of $C_N(c_N; \widehat{\xi})$ and the attainment of suprema used above follow from the continuity hypothesis, under which $\theta \mapsto Q_{j,b}(\theta, \xi)$ is continuous for each fixed (j, b, ξ) . Under (A.14) instead, the argument runs with the word “compact” deleted from the definition of $\mathcal{K}_N$, and in Steps 2 and 3 the statistics $\underline{T}_{j,b,\xi_0}$ and $\overline{T}_{j,b,\xi_0}$ are replaced by the brackets

$$(\sqrt{T_{b,j}} - \Delta_{b,j})_+^2 \leq \underline{T}_{j,b,\xi_0} \leq \overline{T}_{j,b,\xi_0} \leq (\sqrt{T_{b,j}} + \Delta_{b,j})^2,$$

where $T_{b,j}$ and $\Delta_{b,j}$ are T_b and Δ_b of (A.14) computed on block j . The brackets hold because $\mathcal{K}_N \subseteq \mathcal{S}_N(C_0)$ and $\Theta_I^{C_0 \epsilon_N} \in \mathcal{S}_N(C_0)$, they are i.i.d. across blocks, and both converge in law to $\mathcal{T}$ because $T_b \Rightarrow \mathcal{T}$ and $\Delta_b \rightarrow_p 0$.

Step 2: concentration for block subsampling. Partition the sample into $B_N := \lfloor N/b \rfloor$ disjoint blocks of size b , and let block j define $\underline{T}_{j,b,\xi_0}$ and $\overline{T}_{j,b,\xi_0}$. Both statistics are functions of block j alone: $\mathcal{K}_N$ depends only on Θ_I and on the deterministic radius $C_0 \epsilon_N$, and ξ_0 is non-random. Since the blocks are disjoint and the data are i.i.d., $(\underline{T}_{j,b,\xi_0})_{j \leq B_N}$ are i.i.d. within the row, and $\underline{G}(x)$ is an average of B_N i.i.d. Bernoulli indicators. Hence, by Chebyshev’s inequality, for each fixed x ,

$$|\underline{G}(x) - \mathbb{E} \underline{G}(x)| = O_P(B_N^{-1/2}) = O_P(\sqrt{b/N}),$$

and similarly for $\overline{G}(x)$. The Dvoretzky–Kiefer–Wolfowitz inequality upgrades this to a bound uniform in x of the same order. These deviations vanish under $b/N \rightarrow 0$ alone. The *estimated* statistics $\widehat{T}_{j,b}$ are dependent across blocks, because $C_N(c_N; \widehat{\xi})$ is built from the full sample. The two-sided bound of Step 1 is what lets us work with the independent brackets instead, and it is why the argument does not require the condition $b = o(\sqrt{N})$ that a bounded-differences bound applied directly to $\widehat{G}$ would force.

Moreover, blocks are identically distributed, hence

$$\mathbb{E} \underline{G}(x) = \mathbb{P}(\underline{T}_{j,b,\xi_0} \leq x), \quad \mathbb{E} \overline{G}(x) = \mathbb{P}(\overline{T}_{j,b,\xi_0} \leq x).$$

Step 3: small- b limit and quantiles. By Assumption A.7, at every continuity point x of the cdf of $\mathcal{T}$,

$$\mathbb{P}(\underline{T}_{j,b,\xi_0} \leq x) - \mathbb{P}(\mathcal{T} \leq x) \rightarrow 0, \quad \mathbb{P}(\overline{T}_{j,b,\xi_0} \leq x) - \mathbb{P}(\mathcal{T} \leq x) \rightarrow 0,$$

as $b \rightarrow \infty$. Combining with Step 2 gives

$$\underline{G}(x) = P(\mathcal{T} \leq x) + o_P(1), \quad \overline{G}(x) = P(\mathcal{T} \leq x) + o_P(1).$$

Using the two-sided bound, $\hat{\pi}_N = o_P(1)$ and the continuity of the cdf of $\mathcal{T}$ (Assumption A.6), for any fixed continuity point x and any fixed $\epsilon > 0$,

$$P(\mathcal{T} \leq x - \epsilon) + o_P(1) \leq \hat{G}(x) \leq P(\mathcal{T} \leq x + \epsilon) + o_P(1).$$

Letting $\epsilon \downarrow 0$ yields $\hat{G}(x) = P(\mathcal{T} \leq x) + o_P(1)$ at continuity points. Write F for the cdf of $\mathcal{T}$ and $q := q_{1-\tau}(\mathcal{T})$. Since $P(\mathcal{T} = 0) = F(0) < 1 - \tau$, $q > 0$, so F is continuous at q by Assumption A.6 and $F(q) = 1 - \tau$. Since F is strictly increasing near q , $F(q - \delta) < 1 - \tau < F(q + \delta)$ for all small $\delta > 0$, and $q \pm \delta$ can be taken to be continuity points. Applying the convergence of $\hat{G}$ at $q \pm \delta$ shows that the $(1 - \tau)$ -quantile $\hat{c}_\tau$ of $\hat{G}$ lies in $[q - \delta, q + \delta]$ with probability approaching one, so $\hat{c}_\tau \rightarrow_p q$. This proves part (1).

Part (2). Coverage fails exactly when some $\theta \in \Theta_I$ is excluded, i.e. on the event $\{\sup_{\theta \in \Theta_I} N Q_N(\theta, \hat{\xi}) > \hat{c}_\tau\}$. Write $\mathcal{C}_N := \sup_{\theta \in \Theta_I} N Q_N(\theta, \xi_0)$ for the infeasible statistic and $\hat{\mathcal{C}}_N := \sup_{\theta \in \Theta_I} N Q_N(\theta, \hat{\xi})$ for its feasible counterpart. By Assumption B.2(b), which Lemma B.3 verifies from the primitive conditions, $|\hat{\mathcal{C}}_N - \mathcal{C}_N| \leq N \sup_{\theta \in \Theta_I} |Q_N(\theta, \hat{\xi}) - Q_N(\theta, \xi_0)| = o_P(1)$, so $\hat{\mathcal{C}}_N = \mathcal{C}_N + o_P(1)$. By Assumption A.6, $\mathcal{C}_N \Rightarrow \mathcal{T}$, hence $\hat{\mathcal{C}}_N \Rightarrow \mathcal{T}$ by Slutsky. By part (1), $\hat{c}_\tau \rightarrow_p q_{1-\tau}(\mathcal{T})$, which is a continuity point of the cdf of $\mathcal{T}$ by hypothesis. Applying Slutsky's theorem to the pair $(\hat{\mathcal{C}}_N, \hat{c}_\tau)$ and using continuity of the limiting cdf at $q_{1-\tau}(\mathcal{T})$,

$$P(\Theta_I \subseteq \mathcal{C}_N(\hat{c}_\tau; \hat{\xi})) = P(\hat{\mathcal{C}}_N \leq \hat{c}_\tau) \rightarrow P(\mathcal{T} \leq q_{1-\tau}(\mathcal{T})) = 1 - \tau,$$

which is the claim. The procedure is asymptotically exact for Θ_I as a whole, and therefore conservative for any individual $\theta \in \Theta_I$. $\blacksquare$

B.4. Verification of the general conditions

This section proves Corollaries 4.1 to 4.3. We verify the conditions of Appendix A.1, namely Assumptions A.1 to A.4, Assumption A.5, and Assumptions A.6 and A.7; the conclusions are then Theorem A.1, Theorem A.2 and Lemma A.1. Throughout the asymptotic analysis, the number of thresholds S and the number of directions L are fixed.

B.4.1. Affine moment systems Suppose $Eg(Z; \theta, \xi_0) = A\theta - c$ for a fixed $L \times d_\theta$ matrix A and vector c , so that $\Theta_I = \{\theta \in \Theta : A\theta \leq c\}$ is a polyhedron. The linear case of Section 3.1.1 and the example of Section 3.2 are of this form.

LEMMA B.4. (IDENTIFIABILITY AND MAJORANT FOR POLYHEDRA) *Let the polyhedron $\{\theta : A\theta \leq c\}$ be non-empty and contained in Θ , so that $\Theta_I = \{\theta \in \Theta : A\theta \leq c\} = \{\theta : A\theta \leq c\}$. Then Assumption A.1 holds with $C_{\min} = 1/H(A)$ and any $\delta_{\min} > 0$, where $H(A) < \infty$ is the Hoffman constant of A , and Assumption A.5 holds with $C_{\max} = \|A\|$.*

Proof: Hoffman's error bound (Hoffman, 1952) states that for a consistent system $A\theta \leq c$ there is a finite constant $H(A)$, depending on A alone, with $d(\theta, \Theta_I) \leq H(A) \|(A\theta - c)_+\|$ for every θ . Since $\|Eg(Z; \theta, \xi_0)\|_+ = \|(A\theta - c)_+\|$, this is Assumption A.1 with $C_{\min} =$

$1/H(A)$. For the upper bound, let θ^* be the projection of θ on Θ_I . Then $A\theta^* - c \leq 0$, so $\|(A\theta^* - c)_+\| = 0$, and since $v \mapsto \|v\|_+$ is 1-Lipschitz,

$$\|(A\theta - c)_+\| = \left| \|(A\theta - c)_+\| - \|(A\theta^* - c)_+\| \right| \leq \|A(\theta - \theta^*)\| \leq \|A\| d(\theta, \Theta_I).$$

■

For the remaining conditions we additionally use that, in these two cases, $g(Z; \theta, \xi)$ is affine in θ for every fixed ξ . The polyhedron is contained in Θ in both cases. In the linear case, $\ell_{0,k} \geq 0$ and $u_{0,k} \leq 1$ place the polytope inside $\Theta = [0, 1]^S$. In the second example, the grid of directions is symmetric about the origin, so the polyhedron cut out by the L half-spaces is bounded, and Θ is any compact set containing it. For a finite system of affine moment inequalities the polynomial-minorant condition holds automatically, and the constants are computable from A . In the scalar case, $S = 1$, the system is $\ell_0 \leq \theta \leq u_0$ with $A = (-1, 1)'$, so $H(A) = 1$ and $\|A\| = \sqrt{2}$, and Assumption A.1 and Assumption A.5 hold with $C_{\min} = 1$ and $C_{\max} = \sqrt{2}$.

Assumption A.2 is immediate in the affine case. Write $g(Z; \theta, \xi_0) = \beta(Z) + B(Z)\theta$. The class $\{g(\cdot; \theta, \xi_0) : \theta \in \Theta\}$ is contained in a finite-dimensional vector space of measurable functions, hence it is a VC-subgraph of index at most $d_\theta + 2$, and it has a square-integrable envelope $|\beta(Z)| + \|B(Z)\| \sup_{\theta \in \Theta} \|\theta\|$ whenever β and B are square-integrable and Θ is bounded. Such a class is P -Donsker (van der Vaart and Wellner, 1996, Theorem 2.6.7 and Section 2.5). The same two facts give the entropy bound of Assumption A.4(b) with $v = O(d_\theta)$.

Assumptions A.6 and A.7 also follow in the affine case. Because g is affine in θ , the Gaussian limit is $\mathbb{G}(\theta) = \Gamma_0 + \Gamma_1\theta$ for a Gaussian pair (Γ_0, Γ_1) . The population moment $A\theta - c$ is Lipschitz in θ , so Lemma B.5 gives the convergence $\mathcal{C}_N \Rightarrow \mathcal{T}$, the continuity of the cdf of $\mathcal{T}$ on $(0, \infty)$ and (A.14), hence Assumption A.7 for data-dependent sets. Lemma B.5 also gives the strict monotonicity of the cdf of $\mathcal{T}$ on $(0, \infty)$ that Theorem A.2 requires, whenever $P(\mathcal{T} > 0) > 0$.

Finally, (A.8) holds in the affine case whenever Θ_I has non-empty interior. If $B(\theta, \epsilon) \subseteq \Theta_I$ then $B(\theta, \epsilon)$ lies in each half-space $\{\theta : A_l\theta \leq c_l\}$, so θ is at distance at least ϵ from each bounding hyperplane and $c_l - A_l\theta \geq \epsilon\|A_l\|$ for every l and the first part of (A.8) holds with $C = \min_l \|A_l\| > 0$. For the second part, let $B(\theta_c, r) \subset \Theta_I$. For $\epsilon \leq r$, convexity gives $(1 - \epsilon/r)\Theta_I + (\epsilon/r)\theta_c \subset \Theta_I^{-\epsilon}$, so $d_H(\Theta_I^{-\epsilon}, \Theta_I) \leq M\epsilon$ with $M = \text{diam}(\Theta_I)/r$. Lemma A.1 then applies with $\delta = r$.

B.4.2. Moment systems that are not affine Lemma B.5 obtains Assumption A.6 and the uniform approximability (A.14) for moments whose mean is Lipschitz in θ . This covers the affine systems of Appendix B.4.1 and moments that are not continuous in θ , such as (3.12). Steps 1 and 2 of the proof give a self-contained version of the limit theory that Chernozhukov et al. (2007, Section 4) develop for moment-condition models. Write $m(\theta) := E g(Z; \theta, \xi_0)$, $\mathbb{G}_N(\theta) := \mathbb{G}_N g(Z_i; \theta, \xi_0)$, and, for $z \in \ell^\infty(\Theta)^L$,

$$\psi(z) := \sup_{\theta \in \Theta_I} \|I(\theta) + z(\theta)\|_+,$$

so that $\mathcal{T} = \psi(\mathbb{G})^2$ and $N Q_N(\theta, \xi_0) = \|\sqrt{N} m(\theta) + \mathbb{G}_N(\theta)\|_+^2$.

LEMMA B.5. (LIMIT LAW FOR LIPSCHITZ MOMENTS) *Suppose Assumptions A.1 and A.2 hold, m is Lipschitz on Θ with constant L_m and let $b = b_N \rightarrow \infty$ satisfy*

$b(1 \vee c_N)/N \rightarrow 0$. Then Assumption A.6 and (A.14) hold, and the cdf of $\mathcal{T}$ is strictly increasing on $(0, \infty)$ whenever $P(\mathcal{T} > 0) > 0$.

Proof: The proof uses three properties of $v \mapsto \|v\|_+$: it is non-decreasing in each coordinate, it is 1-Lipschitz, and $\|v\|_+ = \sup\{u'v : u \geq 0, \|u\| \leq 1\}$. By the first two, $|\psi(z) - \psi(z')| \leq \sup_{\theta \in \Theta} \|z(\theta) - z'(\theta)\|$, so ψ is continuous on $\ell^\infty(\Theta)^L$. The set Θ_I is non-empty by Assumption A.1 and compact because m is continuous.

Step 1: convergence of $\mathcal{C}_N$. For $\theta \in \Theta_I$, the coordinates with $I_l(\theta) = 0$ have $\sqrt{N}m_l(\theta) = 0$ and the others have a non-negative positive part, so monotonicity gives $\mathcal{C}_N \geq \psi(\mathbb{G}_N)^2$. For $\eta > 0$, let ψ_η be defined as ψ with $I_l(\theta)$ replaced by 0 if $m_l(\theta) > -\eta$ and by $-\infty$ otherwise. On the event $\{\sup_{\theta \in \Theta} \|\mathbb{G}_N(\theta)\| < \sqrt{N}\eta\}$, whose probability tends to one, every coordinate with $m_l(\theta) \leq -\eta$ has $\sqrt{N}m_l(\theta) + \mathbb{G}_{N,l}(\theta) < 0$, and every other coordinate has $\sqrt{N}m_l(\theta) \leq 0$ on Θ_I , so $\mathcal{C}_N \leq \psi_\eta(\mathbb{G}_N)^2$ on that event. By Assumption A.2 and the continuous mapping theorem, $\psi(\mathbb{G}_N) \Rightarrow \psi(\mathbb{G})$ and $\psi_\eta(\mathbb{G}_N) \Rightarrow \psi_\eta(\mathbb{G})$.

As $\eta \downarrow 0$, $\psi_\eta(z)$ decreases to $\psi(z)$ for every z with continuous paths. To see this, take $\eta_n \downarrow 0$ and $\theta_n \in \Theta_I$ that attain $\psi_{\eta_n}(z)$ up to $1/n$. Along a subsequence, $\theta_n \rightarrow \theta^* \in \Theta_I$ and the set $\mathcal{L}$ of coordinates with $m_l(\theta_n) > -\eta_n$ does not depend on n . For $l \in \mathcal{L}$, continuity of m gives $m_l(\theta^*) \geq 0$, hence $m_l(\theta^*) = 0$ and $I_l(\theta^*) = 0$, and continuity of z gives $\lim_n \psi_{\eta_n}(z) \leq \|z_{\mathcal{L}}(\theta^*)\|_+ \leq \psi(z)$, where $z_{\mathcal{L}}$ keeps the coordinates in $\mathcal{L}$. Since $\mathbb{G}$ has continuous paths, $P(\psi_\eta(\mathbb{G})^2 \geq x) \downarrow P(\mathcal{T} \geq x)$ as $\eta \downarrow 0$. At a continuity point x of the cdf of $\mathcal{T}$, the portmanteau theorem and the two bounds give

$$P(\mathcal{T} > x) \leq \liminf_N P(\mathcal{C}_N > x) \leq \limsup_N P(\mathcal{C}_N > x) \leq P(\psi_\eta(\mathbb{G})^2 \geq x)$$

for every $\eta > 0$, and letting $\eta \downarrow 0$ gives $\mathcal{C}_N \Rightarrow \mathcal{T}$.

Step 2: the cdf of $\mathcal{T}$. By the third property, $\psi(\mathbb{G}) = \sup\{u'\mathbb{G}(\theta) : (\theta, u) \in \mathcal{U}\}$, where $\mathcal{U}$ collects the pairs with $\theta \in \Theta_I$, $u \geq 0$, $\|u\| \leq 1$ and $u_l = 0$ whenever $I_l(\theta) = -\infty$. The set $\mathcal{U}$ is compact because m is continuous, so $\psi(\mathbb{G})$ is the supremum of an almost surely bounded, separable, centred Gaussian process, and it is non-negative because $u = 0$ is allowed. The distribution of such a supremum has no atom except possibly at the left end of its support (Tsirel'son, 1975). Since $\mathbb{G}$ has continuous paths on the compact set Θ , its law is a centred Gaussian measure on the separable space $C(\Theta)^L$, whose support is a closed linear subspace and therefore contains the origin. The left end of the support of $\psi(\mathbb{G})$ is therefore 0, because ψ is continuous with $\psi(0) = 0$, and the cdf of $\mathcal{T}$ is continuous on $(0, \infty)$. For strict monotonicity, suppose $P(\mathcal{T} > 0) > 0$. The support of the law of $\mathbb{G}$ then contains some z_0 with $\psi(z_0) > 0$, and it contains tz_0 for every $t \geq 0$ because it is a linear space. Since $\psi(tz_0) = t\psi(z_0)$, for all $x, h > 0$ the open set $\{z : x < \psi(z)^2 < x + h\}$ contains a point of the support and therefore has positive probability.

Step 3: uniform approximability. The observations of a block are i.i.d., so Step 1 at sample size b gives $T_b \Rightarrow \mathcal{T}$. Let $\mathbb{G}_b$ be the empirical process of the block and $\omega_b(r) := \sup_{\|\theta - \theta'\| \leq r} \|\mathbb{G}_b(\theta) - \mathbb{G}_b(\theta')\|$. Because Θ is compact and the paths of $\mathbb{G}$ are continuous, hence uniformly continuous, Assumption A.2 makes $\mathbb{G}_b$ asymptotically uniformly equicontinuous in the Euclidean metric, so $\omega_b(r_b) \rightarrow 0$ in probability for every $r_b \rightarrow 0$ (van der Vaart and Wellner, 1996, Theorem 1.5.7). Let $S \in \mathcal{S}_N(C_0)$. Every $\theta \in S$ has a $\theta' \in \Theta_I$ with $\|\theta - \theta'\| \leq 2C_0\epsilon_N$, and the first two properties of $\|\cdot\|_+$ give

$$\sqrt{bQ_{j,b}(\theta, \xi_0)} \leq \sqrt{bQ_{j,b}(\theta', \xi_0)} + 2\sqrt{b}L_mC_0\epsilon_N + \omega_b(2C_0\epsilon_N).$$

The same bound holds with the roles of S and Θ_I exchanged. Hence (A.14) holds with $\Delta_b := 2\sqrt{b} L_m C_0 \epsilon_N + \omega_b(2C_0 \epsilon_N)$, the same function of the data of each block, and $\Delta_b = o_P(1)$ because $b \epsilon_N^2 = b(1 \vee c_N)/N \rightarrow 0$ and $\omega_b(2C_0 \epsilon_N) = o_P(1)$. ■

B.4.3. Proof of Corollary 4.1 In this example, A collects the $2S$ coordinate moment functions (3.4) and the $S - 1$ increment moment functions (3.5), and $\theta \mapsto E g(Z; \theta, \xi)$ is affine with $B(Z)$ constant. Lemma B.4 and the preceding paragraphs give Assumption A.1, A.2, A.5 and Assumption A.4(b). Let

$$\Xi_N := \left\{ \xi = (p, F^1(w_1 | \cdot), \dots, F^1(w_S | \cdot)) : p \geq \underline{p}, \|p - p_0\|_{P,2} \leq g_N^p, \right. \\ \left. \max_{j \leq S} \|F^1(w_j | \cdot) - F_0^1(w_j | \cdot)\|_{P,2} \leq g_N^F \right\}.$$

Then Assumption A.3(a) holds since $\xi_0 \in \Xi_N$, (b) holds with $\phi_N \rightarrow 0$ by Assumptions 4.2 and 4.3, and (c) holds with $g_N := g_N^p + \sqrt{S} g_N^F = o(1)$.

Orthogonality and the second-order remainder. Only the S lower-bound moment functions involve the nuisance $\xi = (p, F^1(w_1 | \cdot), \dots, F^1(w_S | \cdot))$. Fix a grid point w and write $\xi_0 = (p_0, F_0^1(w | \cdot))$ for the true nuisance, and define deviations $\delta_p(x) := p(x) - p_0(x)$ and $\delta_F(x) := F^1(w | x) - F_0^1(w | x)$. Because $E[D | X] = p_0(X)$ and $E[D \mathbf{1}\{W \leq w\} | X] = p_0(X) F_0^1(w | X)$,

$$m(\xi) := E \phi_L(Z; w, \xi) = E \left[F^1(w | X) + \frac{p_0(X)}{p(X)} (F_0^1(w | X) - F^1(w | X)) \right].$$

Substituting $\xi_r = \xi_0 + r(\xi - \xi_0)$ and rearranging gives

$$m(\xi_r) - m(\xi_0) = r^2 E \left[\frac{\delta_F(X) \delta_p(X)}{p_0(X) + r \delta_p(X)} \right] \quad \text{for every } r. \quad (\text{B.13})$$

Assumption A.3(d) follows by differentiating (B.13) at $r = 0$. Assumption A.3(e) follows from (B.13) at $r = 1$: since $p_0 \geq \underline{p}$ and $p \geq \underline{p}$ on Ξ_N by Assumption 4.2, the denominator satisfies $p_0 + r \delta_p \geq \underline{p}$, and Cauchy-Schwarz gives

$$|m(\xi) - m(\xi_0)| \leq \underline{p}^{-1} \|\delta_p(X)\|_{P,2} \|\delta_F(X)\|_{P,2}, \quad (\text{B.14})$$

so $\bar{s}_N$ can be taken as the product of the two first-stage rates,

$$\bar{s}_N = \underline{p}^{-1} g_N^p g_N^F = o(N^{-1/2}). \quad (\text{B.15})$$

Envelope and continuity in the nuisance. Under the overlap condition, $|\phi_L| \leq 1 + \underline{p}^{-1}$ and $|\phi_U| \leq 1$, so the envelope of Assumption A.4(a) may be taken constant with a finite c -norm for every c . For Assumption A.4(c), ϕ_L is Lipschitz in ξ , and $|\phi_L(\xi) - \phi_L(\xi_0)| \leq (1 + \underline{p}^{-1}) |\delta_F(X)| + \underline{p}^{-2} |\delta_p(X)|$, so $r'_N \leq (1 + \underline{p}^{-1} + \underline{p}^{-2}) g_N$. Then $r'_N \sqrt{\log N} = o(1)$ whenever $g_N = o((\log N)^{-1/2})$, which is attained by any polynomial rate.

Therefore, all the conditions of Appendix A.1 hold, and Theorem A.1, Theorem A.2 and Lemma A.1 deliver conclusions (1), (2) and (3) of Corollary 4.1.

B.4.4. Proof of Corollary 4.2 For $\theta \in \Theta$ the population moments are

$$m(\theta) = \left(F_0^{\text{lo}}(\theta_1) - \alpha_1, F_0^{\text{lo}}(\theta_2) - \alpha_2, \alpha_1 - F_0^{\text{up}}(\theta_1), \right. \\ \left. \alpha_2 - F_0^{\text{up}}(\theta_2), F_0^{\text{up}}(\theta_2) - F_0^{\text{up}}(\theta_1) - (\alpha_2 - \alpha_1) \right)'.$$

The densities of F_0^{lo} and F_0^{up} are $\mathbb{E}[f_0^1(w \mid X)]$ and $\mathbb{E}[p_0(X)f_0^1(w \mid X)]$. By Assumption 4.4(2) both are at most $\bar{f}$, so m is Lipschitz with constant $L_m \leq 3\bar{f}$. By Assumption 4.4(3), and since $p_0 \leq 1$, both are at least $\underline{f}$ almost everywhere on $\mathcal{J}$, so $|F(t) - F(t')| \geq \underline{f}|t - t'|$ for $t, t' \in \mathcal{J}$ and $F \in \{F_0^{\text{lo}}, F_0^{\text{up}}\}$. The set Θ_I contains θ_0 by the derivation of (3.11).

Identifiability and majorant. Assumption A.5 holds with $C_{\max} = L_m$, by the argument in the proof of Lemma B.4 with $A\theta - c$ replaced by $m(\theta)$. For Assumption A.1, let $\Phi(\theta) := (F_0^{\text{up}}(\theta_1), F_0^{\text{up}}(\theta_2))$. The density bounds make Φ one-to-one from $\mathcal{J}^2$ onto $\Phi(\mathcal{J})^2$, with $\|\Phi(\theta) - \Phi(\theta')\| \leq \bar{f}\|\theta - \theta'\|$ on Θ and $\|\theta - \theta'\| \leq \underline{f}^{-1}\|\Phi(\theta) - \Phi(\theta')\|$ on $\mathcal{J}^2$. With $\bar{s}_j := F_0^{\text{up}}(\bar{\theta}_j)$, consider the polygon

$$\mathcal{R} := \{s \in \mathbb{R}^2 : \alpha_j \leq s_j \leq \bar{s}_j, j = 1, 2, \quad s_2 - s_1 \leq \alpha_2 - \alpha_1, \quad s_1 \leq s_2\} = \{s : As \leq c\}.$$

Every $\theta \in \Theta_I$ satisfies $\theta_j \leq \bar{\theta}_j$, so $\Theta_I \subset \mathcal{J}^2$. Since F_0^{lo} and F_0^{up} are strictly increasing on $\mathcal{J}$, it follows that $\mathcal{R} \subset \Phi(\mathcal{J})^2$ and $\Theta_I = \{\theta \in \mathcal{J}^2 : \Phi(\theta) \in \mathcal{R}\}$.

Fix $\theta \in \Theta$ with $d(\theta, \Theta_I) \leq \bar{\delta}$, so that $\theta \in \mathcal{J}^2$, and let $s := \Phi(\theta)$. The positive parts of the rows $s_j \geq \alpha_j$ and $s_2 - s_1 \leq \alpha_2 - \alpha_1$ are those of the third to fifth coordinates of $m(\theta)$, and the row $s_1 \leq s_2$ holds because $\theta_1 \leq \theta_2$. The density bounds on $\mathcal{J}$ give $(s_j - \bar{s}_j)_+ \leq \bar{f}(\theta_j - \bar{\theta}_j)_+ \leq (\bar{f}/\underline{f})(F_0^{\text{lo}}(\theta_j) - \alpha_j)_+$. Hence $\|(As - c)_+\| \leq (\bar{f}/\underline{f})\|m(\theta)\|_+$. Hoffman's bound gives $s^* \in \mathcal{R}$ with $\|s - s^*\| \leq H(A)\|(As - c)_+\|$, and $\theta^* := \Phi^{-1}(s^*) \in \Theta_I$ satisfies $\|\theta - \theta^*\| \leq \underline{f}^{-1}\|s - s^*\|$, so $d(\theta, \Theta_I) \leq H(A)\bar{f}\underline{f}^{-2}\|m(\theta)\|_+$. On the compact set $\{\theta \in \Theta : d(\theta, \Theta_I) \geq \bar{\delta}\}$ the continuous function $\theta \mapsto \|m(\theta)\|_+$ is positive and, if the set is non-empty, attains a minimum $\kappa_{\bar{\delta}} > 0$; otherwise set $\kappa_{\bar{\delta}} := \bar{\delta}$. Assumption A.1 therefore holds with $\delta_{\min} = \bar{\delta}$ and $C_{\min} = \min\{\underline{f}^2/(H(A)\bar{f}), \kappa_{\bar{\delta}}/\bar{\delta}\}$.

Donsker property and entropy. For $\xi = (p, F^1)$ with $F^1(\cdot \mid x)$ non-decreasing, the subgraphs of the class $\{x \mapsto F^1(t \mid x) : t \in [\underline{w}, \bar{w}]\}$ are totally ordered by inclusion, so the class is VC-subgraph with index 2. The same holds for $\{D \mathbf{1}\{W \leq t\} : t \in \mathbb{R}\}$, which depends on W only through DW , and $\{D \mathbf{1}\{t_1 < W \leq t_2\} : t_1 \leq t_2\}$ is VC-subgraph with index at most 3. Each coordinate of $g(\cdot; \theta, \xi)$ is a fixed bounded function plus at most two such functions, each multiplied by one of the fixed functions $1 - D/p(X)$, $1/p(X)$ and 1 , which are bounded by $1 + \underline{p}^{-1}$. Multiplication by a fixed function preserves the VC-subgraph property (van der Vaart and Wellner, 1996, Lemma 2.6.18), and the uniform covering number of a sum of two classes is at most the product of theirs, so Assumption A.4(b) holds with constants that depend only on $\underline{p}$, and the envelope is the constant $2 + \underline{p}^{-1}$. A class with a bounded envelope and a uniform entropy bound of the form (A.4) is $\bar{P}$ -Donsker (van der Vaart and Wellner, 1996, Theorem 2.5.2). By Assumption 4.4(2), $\mathbb{E}|F_0^1(t \mid X) - F_0^1(t' \mid X)| \leq \bar{f}|t - t'|$ and $\mathbb{P}(D = 1, t < W \leq t') \leq \bar{f}|t - t'|$ for $t < t'$, so $\mathbb{E}\|g(Z; \theta, \xi_0) - g(Z; \theta', \xi_0)\|^2 \leq C\|\theta - \theta'\|$ with C depending on $(\underline{p}, \bar{f})$. The Gaussian limit has paths that are uniformly continuous in this L^2 semimetric (van der Vaart and Wellner, 1996, Section 1.5), and hence continuous in θ , which completes Assumption A.2.

First stage. Measure the nuisance error by $\|p - p_0\|_{P,2} + \sup_w \|F^1(w \mid \cdot) - F_0^1(w \mid \cdot)\|_{P,2}$, with the supremum over $[\underline{w}, \bar{w}]$, and let

$$\Xi_N := \left\{ \xi = (p, F^1) : p \geq \underline{p}, F^1(\cdot \mid x) \text{ non-decreasing with values in } [0, 1] \forall x, \right. \\ \left. \|p - p_0\|_{P,2} \leq g_N^p, \sup_w \|F^1(w \mid \cdot) - F_0^1(w \mid \cdot)\|_{P,2} \leq g_N^F \right\}.$$

Then Assumption A.3(a)–(c) hold as in the proof of Corollary 4.1, now by Assumptions 4.2 and 4.5, with $g_N := g_N^p + g_N^F$. Only the two lower-bound moments involve ξ , and they evaluate ϕ_L at $w = \theta_j$. The identity (B.13) holds at every threshold w , so Assumption A.3(d) holds at every θ , and (B.14) at $w = \theta_j$, with $\|F^1(w | \cdot) - F_0^1(w | \cdot)\|_{P,2} \leq g_N^F$ for every w , gives Assumption A.3(e) with $\bar{s}_N \leq \sqrt{2} \underline{p}^{-1} g_N^p g_N^F = o(N^{-1/2})$, uniformly in $\theta \in \Theta$. The envelope and continuity bounds for ϕ_L in the proof of Corollary 4.1 hold at every threshold, so Assumption A.4(a) and (c) hold with $r'_N \leq \sqrt{2} (1 + \underline{p}^{-1} + \underline{p}^{-2}) g_N$.

Limit law and degeneracy. Lemma B.5 gives Assumption A.6 and (A.14), which replaces the continuity hypothesis and Assumption A.7 in Theorem A.2. For conclusion (3), suppose Θ_I has non-empty interior; then so does $\mathcal{R}$, which contains a ball $B(s_c, r)$. Let $\delta' := \min\{\bar{\delta}, r/\bar{f}\}$, let $0 < \epsilon \leq \delta'$ and let $B(\theta, \epsilon) \subseteq \Theta_I$. The points $\theta \pm \epsilon e_j$ and $\theta + (\epsilon/\sqrt{2})(-1, 1)'$ lie in Θ_I , so $\theta_j + \epsilon \leq \theta_j$, $\theta_j - \epsilon \geq \underline{\theta}_j$ and $F_0^{\text{up}}(\theta_2 + \epsilon/\sqrt{2}) - F_0^{\text{up}}(\theta_1 - \epsilon/\sqrt{2}) \leq \alpha_2 - \alpha_1$, and the density bounds on $\mathcal{J}$ give $m_l(\theta) \leq -\underline{f}\epsilon$ for every l . For the second part of (A.8), let $\epsilon' := \bar{f}\epsilon \leq r$. If $B(s', \epsilon') \subseteq \mathcal{R}$, then $\theta' := \Phi^{-1}(s') \in \Theta_I$ satisfies $B(\theta', \epsilon) \subseteq \mathcal{J}^2$ and $\Phi(B(\theta', \epsilon)) \subseteq B(s', \epsilon')$, hence $B(\theta', \epsilon) \subseteq \Theta_I$; in particular $\Phi^{-1}(s_c) \in \Theta_I^{-\epsilon}$. The convexity argument of Appendix B.4.1 places every point of $\mathcal{R}$ within $\epsilon' \text{diam}(\mathcal{R})/r$ of such an s' , and pulling back with Φ^{-1} gives $d_H(\Theta_I^{-\epsilon}, \Theta_I) \leq \bar{f} \text{diam}(\mathcal{R}) \epsilon / (\underline{f} r)$. Lemma A.1 then applies with $\delta = \delta'$, $C = \underline{f}$ and $M = \bar{f} \text{diam}(\mathcal{R}) / (\underline{f} r)$.

All conditions of Appendix A.1 therefore hold, with (A.14) in place of continuity in θ and Assumption A.7, and Theorem A.1, Theorem A.2 and Lemma A.1 deliver conclusions (1), (2) and (3) of Corollary 4.2.

B.4.5. Proof of Corollary 4.3 Write $V_0 := V(\eta_0)$ and $\bar{\theta} := \sup_{\theta \in \Theta} \|\theta\|$. At the true nuisance, $E g_l(Z; \theta, \xi_0) = q_l' \Sigma_0 \theta - c_l$, where c_l is the right-hand side of (3.18), so the system is affine with the $L \times d$ matrix A whose rows are $q_l' \Sigma_0$. Since Σ_0 is non-singular by Assumption 4.6(1) and the symmetric grid spans $\mathbb{R}^d$ by Assumption 4.6(2), the rows $q_l' \Sigma_0$ positively span $\mathbb{R}^d$, so the polyhedron $\{\theta : A\theta \leq c\}$ is bounded. It is non-empty, since it contains $\Theta_I^* \ni \theta_0$, and Θ contains it by Assumption 4.6(3), so Lemma B.4 applies. For each fixed ξ , $g(Z; \theta, \xi)$ is affine in θ with bounded coefficients, so Appendix B.4.1 gives Assumptions A.1, A.2 and A.5, Assumption A.4(b), Assumptions A.6 and A.7, and (A.8) whenever Θ_I has non-empty interior. Let

$$\Xi_N := \left\{ \xi = (\eta, \ell, \pi_1, \dots, \pi_L) : \|\eta(X)\| \leq \bar{D}, |\ell(X)| \leq \bar{Y}, 0 \leq \pi_l(X) \leq \bar{\Delta} \text{ a.s.,} \right. \\ \left. \|\eta - \eta_0\|_{P,2} \leq g_{\eta,N}, \|\ell - \ell_0\|_{P,2} \leq g_{\ell,N}, \right. \\ \left. \max_{l \leq L} \|\pi_l - \pi_{0,l}\|_{P,2} \leq g_{\pi,N} \right\}.$$

By Assumption 4.7, $\eta_0 = E[D | X]$, $\ell_0 = E[Y_L | X]$ and $\pi_{0,l} = E[\Delta \mathbf{1}\{q_l' V_0 > 0\} | X]$ satisfy the same bounds, so Assumption A.3(a) holds; (b) holds by Assumption 4.9; and (c) holds with $g_N := g_{\eta,N} + g_{\ell,N} + \sqrt{L} g_{\pi,N} = o(1)$.

Orthogonality and the second-order remainder. For $\xi \in \Xi_N$ write $\delta := \eta - \eta_0$, $\lambda := \ell - \ell_0$ and $\nu_l := \pi_l - \pi_{0,l}$, so that $V(\eta) = V_0 - \delta(X)$, and let $m_l(\theta, \xi) := E g_l(Z; \theta, \xi)$. Conditioning on X and using $E[V_0 | X] = 0$ and $E[Y_L - \ell_0(X) | X] = 0$, the first two terms of (3.20) contribute $E[(q_l' \delta)(\delta' \theta)] - E[(q_l' \delta) \lambda]$ to $m_l(\theta, \xi) - m_l(\theta, \xi_0)$. For the bracketed term, $E[(q_l' V(\eta)) \pi_l] = -E[(q_l' \delta) \pi_{0,l}] - E[(q_l' \delta) \nu_l]$, and the definition of $\mathcal{K}_l$ together with

$E[\Delta \mathbf{1}\{q'_l V_0 > 0\} \mid X] = \pi_{0,l}(X)$ gives $E[\Delta(q'_l V(\eta))_+] - E[\Delta(q'_l V_0)_+] = \mathcal{K}_l(\eta) - E[(q'_l \delta)\pi_{0,l}]$. Hence

$$\begin{aligned} m_l(\theta, \xi) - m_l(\theta, \xi_0) &= E[(q'_l \delta(X))(\delta(X)' \theta)] - E[(q'_l \delta(X))\lambda(X)] \\ &\quad - E[(q'_l \delta(X))\nu_l(X)] - \mathcal{K}_l(\eta). \end{aligned} \quad (\text{B.16})$$

Along $\xi_r = \xi_0 + r(\xi - \xi_0)$ the deviations are $r\delta$, $r\lambda$ and $r\nu_l$, and $\eta_0 + r\delta = (1-r)\eta_0 + r\eta$ satisfies $\|\eta_0 + r\delta\| \leq \bar{D}$, so Assumption 4.8 gives $|\mathcal{K}_l(\eta_0 + r\delta)| \leq C_K r^2 \|\delta\|_{P,2}^2$. Every term of (B.16) along the path is therefore $O(r^2)$, and differentiating at $r = 0$ gives Assumption A.3(d). At $r = 1$, the Cauchy–Schwarz inequality, $\|q_l\| = 1$ and Assumption 4.8 give

$$\sup_{\theta \in \Theta} |m_l(\theta, \xi) - m_l(\theta, \xi_0)| \leq (\bar{\theta} + C_K) \|\delta\|_{P,2}^2 + \|\delta\|_{P,2} \|\lambda\|_{P,2} + \|\delta\|_{P,2} \|\nu_l\|_{P,2},$$

so Assumption A.3(e) holds with $\bar{s}_N \lesssim g_{\eta,N}(g_{\eta,N} + g_{\ell,N} + g_{\pi,N}) = o(N^{-1/2})$ by Assumption 4.9.

Envelope and continuity in the nuisance. For $\xi \in \Xi_N \cup \{\xi_0\}$, Assumption 4.7 and the bounds in Ξ_N give $\|V(\eta)\| \leq 2\bar{D}$, $|Y_L - \ell(X)| \leq 2\bar{Y}$, $0 \leq \pi_l \leq \bar{\Delta}$ and $\Delta \leq \bar{\Delta}$, so every coordinate of g is bounded by a constant that depends only on $(\bar{D}, \bar{Y}, \bar{\Delta}, \bar{\theta})$, and Assumption A.4(a) holds for every $c > 2$. Each of the four terms of (3.20) is Lipschitz in $(\delta(X), \lambda(X), \nu_l(X))$ with such a constant, because $a \mapsto a_+$ is 1-Lipschitz. Hence

$$\sup_{\theta \in \Theta} (E\|g(Z; \theta, \xi) - g(Z; \theta, \xi_0)\|^2)^{1/2} \lesssim \|\delta\|_{P,2} + \|\lambda\|_{P,2} + \max_{l \leq L} \|\nu_l\|_{P,2},$$

so Assumption A.4(c) holds with $r'_N \lesssim g_{\eta,N} + g_{\ell,N} + g_{\pi,N}$, and $r'_N \sqrt{\log N} = o(1)$ by Assumption 4.9.

All the conditions of Appendix A.1 therefore hold, and Theorem A.1, Theorem A.2 and Lemma A.1 deliver conclusions (1), (2) and (3) of Corollary 4.3.

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